



文学硕士学位论文

灾害事故报道的多模态话语分析—以 China Daily 和 BBC News 为例

Multimodal Discourse Analysis of Disasters and Accidents Coverage-Taking China Daily and BBC News as Examples

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摘 要

全球范围内灾害事故频发,对人们的生命财产安全造成极大威胁。媒体通过新闻报道向社会公布灾害成因、影响范围、造成的损失等情况,满足公众的信息需求。滞后、不恰当、不准确的新闻破坏了媒体的专业性和社会服务性。因此,提高灾害新闻内容的准确性和发布的及时性,成为了媒体部门迫切需要解决的问题。新闻报道逐渐融入了图像、视频、音乐、色彩等模态的资源。然而,复杂多样的要素亦会导致信息繁多、主次不清,视觉体验不佳。为了优化符号的组合与布局,让新闻更加生动、直观,本研究提出了以下问题:(1)本语料中语言资源三大元功能的占比及其作用是怎样的?(2)本语料中图像资源三种意义的占比及其作用是怎样的?

本文梳理了灾害新闻的特点和多模态话语分析的研究现状,明确了研究对象、理论和方法,收集了《中国日报》和《英国广播公司新闻》中的120篇自然灾害新闻文本和160幅图像,建立了比较语料库。以系统功能语法和视觉语法构成的多模态话语分析框架为理论依据,研究者使用Uam Corpus Tool和Uam Image Tool软件标注新闻中的文字和图像,借助话语分析法阐释案例中语言和图像在灾害背景下发挥的作用。

研究发现:(1)概念功能与再现意义负责传播灾害信息和科普应急知识。概念功能方面,物质过程、言语过程和关系过程占比较高。再现意义方面,英方报道中叙事再现的占比(67.47%)显著高于其在中方报道中的占比(48.75%)。(2)人际功能与互动意义可以鼓舞受灾群众的信心,激发读者的情感关怀。在人际功能方面,大量的中量值和低量值情态词使新闻报道更加客观真实。祈使语气可以再现事故发生时的紧急状况,虚拟语气常被用来宣传应急措施。在互动意义方面,图像中的接触、距离、视角在不同程度上加强了有关主体的情感互动。中方报道中中量值情态的图像比例(40.26%)略高于其在英方报道中的比例(37.04%)。(3)文本功能与构图意义可以优化要素布局、吸引读者注意力。在文本功能方面,英方和中方报道中无标记主位和主题主位的比例最高,使重点内容更加突出。在构图意义方面,图像采用中心-边缘布局、前景化、强饱和度等策略使受众的目光聚焦于关键信息。中方新闻采用相对大小来突出信息的占比(28.09%)比其在英方新闻中的占比(16.87%)高11.22%

综上所述,本文对比了中英自然灾害新闻中话语和图像在灾害语境中的作用及其占比,丰富了多模态话语分析理论的应用范围并增强了它们对新闻语篇的解释力。这帮助读者了解了中英媒体建构自然灾害事件的话语特征,为优化新闻的话语策略、促进媒体、管理部门、公众进行协同应急管理提供借鉴。

关键词: 灾害事故新闻;多模态话语分析;系统功能语法;视觉语法;应急管理

Abstract

Disasters occur frequently around the world, posing a great threat to people's lives and property. The media informs the public of the causes of disasters, the scope of impact, the losses caused, and other information to meet the public's information needs through news. Delayed, inappropriate, and inaccurate news undermines the professionalism and social service role of the media. Therefore, improving the accuracy of disaster news content and the timeliness of its release has become an urgent issue that the media needs to solve. News reports gradually incorporate resources in various modalities such as images, videos, music, and colors. However, complex and diverse elements can lead to too much information, unclear priorities, and poor visual experience. In order to optimize the combination and layout of symbols and make the news more vivid and intuitive, this study raises the following questions: (1) What are the proportions of the three meta-functions of language resources and their functions in the two corpora? (2) What are the proportions of the three meanings of image resources and their functions in the two corpora?

This thesis sorts out the characteristics of disaster news and the current status of multimodal discourse analysis, clarifies the research objects, theories and methods, collects 120 natural disaster news texts and 160 images from China Daily and BBC News, and establishes comparable corpora. Based on the multimodal discourse analysis framework composed of systemic functional grammar and visual grammar, the researcher uses Uam Corpus Tool and Uam Image Tool software to annotate the text and images in the news, and applies discourse analysis to explain the role of language and images in the disaster context through specific cases.

The study found that: (1) The ideational function and representational meaning are responsible for disseminating disaster information and popularizing emergency knowledge. In terms of ideational function, material process, verbal process and relational process account for a high proportion. In terms of representational meaning, the proportion of narrative representation in British reports (67.47%) is significantly higher than that in Chinese reports (48.75%). (2) The interpersonal function and interactive meaning can inspire the confidence of the victims and stimulate the readers' emotional care. Regarding interpersonal function, a large number of medium-value and low-value modal words make news more objective and truer. The imperative mood can reproduce the emergency situation when the disaster occurs, and the subjunctive mood is usually adopted to promote emergency measures. Regarding interactive meaning, the contact, distance and perspective in the picture strengthen the emotional interaction of the relevant subjects to varying degrees. The proportion of images with medium-value modal in Chinese reports (40.26%) is slightly higher than that in British reports (37.04%). (3) The textual function and compositional meaning can optimize the layout of elements and attract readers' attention.

Regarding textual function, the proportion of unmarked themes and topical themes in the British and Chinese reports is the highest, making the key content more prominent. Regarding compositional meaning, the image adopts strategies such as center-margin layout, foregrounding, and strong saturation to focus the audience's attention on key information. The proportion of Chinese news using relative size to highlight information (28.09%) is 11.22% higher than that in British news (16.87%).

In summary, this paper compares the role of discourse and images in disaster context and their proportions in Chinese and English news coverage, enriches the application scope of multimodal discourse analysis theory and enhances its explanatory power for news discourse. It helps readers understand the discourse characteristics of natural disaster events constructed by Chinese and British media, and provides references for optimizing news discourse strategies and promoting collaborative emergency management among the media, management departments and the public.

Keywords: Disasters and accidents coverage; Multimodal discourse analysis; Systemic functional grammar; Visual grammar; Emergency management

Contents

1.Introduction.....	1
1.1 Research Background	1
1.2 Research Purposes and Significance.....	3
1.3 Thesis Structure.....	4
2.Literature Review	6
2.1 Multimodal Discourse Analysis (MDA).....	6
2.1.1 Definition of multimodal discourse analysis	6
2.1.2 Research on multimodal discourse analysis abroad	6
2.1.3 Research on multimodal discourse analysis at home	8
2.2 Natural Disaster Reporting	9
2.2.1 Definition of natural disasters	10
2.2.2 Research on natural disaster news abroad	10
2.2.3 Research on natural disaster news at home	12
3. Theoretical Basis and Research Design.....	14
3.1 Theoretical Basis	14
3.1.1 Systemic functional grammar	14
3.1.2 Visual grammar	16
3.2 Research Design.....	18
3.2.1 Research methods	18
3.2.2 Research procedure	19
4. Comparative Analysis of Natural Disaster News Language	24
4.1 Ideational Function of News Discourse.....	24
4.1.1 Material process	26
4.1.2 Verbal process	29
4.1.3 Relational process	32
4.2 Interpersonal Function of News Discourse.....	34
4.2.1 Modality system	35
4.2.2 Mood system	37
4.3 Textual Function of News Discourse	40
4.3.1 Theme-rheme structure	40
4.3.2 Information connection and coherence	42
5. Comparative Analysis of Natural Disaster News Images.....	45
5.1 Representational Meaning of Images	45
5.1.1 Conceptual representation.....	46

5.1.2 Narrative representation	47
5.2 Interactive Meaning of Images	49
5.2.1 Contact and distance	50
5.2.2 Perspective and modality	52
5.3 Compositional Meaning of Images	53
5.3.1 Information value and saliency	54
5.3.2 Framing	56
6. Conclusion	58
6.1 Findings and Implications	58
6.2 Limitations and Suggestions	60
References	62

List of Figures

Number	Name of Figures	Pages
Fig. 3.1	The number and types of natural disasters in the collected corpora	20
Fig. 3.2	Operation of Uam Corpus Tool 3.3x	21
Fig. 3.3	Operation of Uam Image Tool 2.1	21
Fig. 3.4	Keyword statistics operation of Antconc 3.5.9	21
Fig. 3.5	Modal word statistics operation of Antconc 3.5.9	22
Fig. 3.6	Analysis framework of natural disaster news	22
Fig. 5.1	An aerial drone photo taken on July 28, 2024, shows rescuers relocating residents at Jingtang village of Zixing city, Central China's Hunan province.	46
Fig. 5.2	Aerial view of the lava flow during the Mount Nyiragongo volcanic eruption near Goma, in the Democratic Republic of Congo May 23, 2021.	46
Fig. 5.3	The world is on the brink of "catastrophic warming". (China Daily, Zheng Huawei, April 10, 2023.	46
Fig. 5.4	People try to extinguish a forest fire in Nemocon, Colombia, on Thursday.	48
Fig. 5.5	Magma comes out from "rifts".	48
Fig. 5.6	Xi urges "all-out effort" to protect lives in flood prevention.	48
Fig. 5.7	Rescuers transfer flood-trapped people in Zhuozhou, Baoding city, North China's Hebei province	52
Fig. 5.8	Rescuers evacuate a baby after a mudslide hit Pingwu county, Sichuan province, on Tuesday.	52
Fig. 5.9	Rescue operations continue in Karatsu.	52
Fig. 5.10	President Chakwera visited some victims.	52
Fig. 5.11	4 killed, 23 missing in Southwest China flash flood, mudslide	52
Fig. 5.12	A close view at the Vistula River as water levels are at record low in Warsaw, Poland.	52
Fig. 5.13	A fire crew member fights part of the Hillville bushfire south of Taree in New South Wales, Australia, on Tuesday.	55
Fig. 5.14	Members of Chongqing Blue Sky Rescue Team carry out the search and rescue mission in quake-hit Malatya	55
Fig. 5.15	Firefighters were helped by wetter weather towards the end of the week.	55
Fig. 5.16	Destroyed houses in pandai sikek, Tanah Datar Regency, West Sumatera	57
Fig. 5.17	Macroseismic Intensity Map	57

List of Tables

Number	Name of Tables	Pages
Table 4.1	Frequency and proportion of processes based on transitivity	24
Table 4.2	Top 20 keywords in China Daily and BBC News	25
Table 4.3	Frequency and proportion of various modal words	35
Table 4.4	Frequency and proportion of thematic categories in natural disaster news	41
Table 5.1	Frequency and proportion of conceptual representation and narrative representation	45
Table 5.2	Frequency and proportion of elements in interactive meaning of images	49
Table 5.3	Frequency and proportion of elements in compositional meaning of images	53

1. Introduction

In the face of various natural disasters, the media disseminates relevant information to people through television, radio, news, etc. Disaster accident news uses multimodal resources such as images, colors, and actions to disseminate information, which has received widespread attention from the society. Moreover, the word collocation, image combination, and overall layout in the news text have a certain impact on the effectiveness of information dissemination. The accuracy of news content, the timeliness of release, and the rationality of graphic and text layout are important criteria for measuring news quality. Problems such as the inconsistency between pictures and texts and cluttered content in news hinder people's acceptance of information.

In the tense and public opinion-filled disaster context, the release of timely and useful information can clarify rumors and appease the uneasy psychology of the masses. Therefore, it is necessary to clarify the dominant and subordinate status and the role of language and image symbols in disaster news, which is very necessary to improve the attractiveness and readability of news. When sudden natural disasters occur, many people become nervous and at a loss due to their lack of emergency response experience, resulting in missing the best opportunity to escape or seek help. As a commonly used channel for obtaining information by the public, news media can popularize safety knowledge to the audience with their strong communication power, thus giving full play to their social value. This study provides references for the news practitioners to provide concise, eye-catching, and vivid emergency rescue propaganda pictures, thereby helping the masses to quickly master rescue and escape skills.

In view of the various classifications and causes of disaster accidents, this study selects natural disaster news as the main research object. This is mainly because natural disasters are caused by natural causes such as geology, precipitation, and high temperature, which are universal in China and the UK, and it is easy to collect a large amount of corpus. Language and images, as meaningful symbolic resources, are skillfully used by news creators according to certain purposes and rules to play the role of prompting, warning, and guiding readers. Under the guidance of systemic functional grammar and visual grammar, the researcher uses corpus tools to count the proportions of symbolic resources, analyzing the role of image symbols and language symbols in disaster context.

1.1 Research Background

The United Nations Office for Disaster Risk Reduction (UNDRR) released the Global Assessment Report (GAR 2022) in 2022, indicating that in the past 20 years, 350 to 500 large and medium-sized disaster events occurred every year. It is estimated that by 2030, the number of large

and medium-sized disasters will reach 560 per year, and the average annual loss caused by global disasters will reach 170 billion US dollars. Disaster events have caused huge losses to people's lives and economic interests, and hindered the development of normal social life. For example, in October 2022, consecutive earthquakes occurred in northwestern Afghanistan, killing more than 1,400 people. In May 2023, heavy rains in eastern Congo (Kinshasa) caused floods, mudslides, and landslides, with economic losses exceeding 10 million US dollars (<http://www.zaihai.cn/>). The situation of disaster prevention and control is very serious both internationally and domestically. The *14th Five-Year Plan for National Comprehensive Disaster Prevention and Reduction* issued by the China Disaster Reduction Committee in 2022 pointed out that under the background of global warming, extreme weather and climate events in China are frequent, and natural disasters such as high temperature, heavy rain, floods, and droughts are prone to occur. When a disaster occurs, the media, as an objective observer, can provide important information to the affected areas and the entire society (Miles & Morse, 2007). A survey finds that 39% of readers read "accidents and disasters" news, while 25% read political news and 33% read general news (Sood et al., 1987).

As the "megaphone" of society, traditional and new media have adapted to the needs of readers of different ages and reading habits, playing a role in disaster risk response. On the one hand, news reports play an important role in disaster warning, material collection, community communication and other tasks. On the other hand, disaster reports are of great significance in building the image of active and capable national workers and enhancing the credibility of the government. This is also an important part of rallying people's hearts and stabilizing social order under the background of disasters. Therefore, natural disaster reports deserve our attention and research.

Discourse analysis is a powerful tool for revealing social problems and, to a certain extent, promotes the reasonable resolution of related events. Some news reports have problems such as missing information, vague and stiff wording, and lack of humanistic care. The media may overemphasize the tragic experiences of disaster victims in their reports, thus causing secondary psychological trauma to them. These are not conducive to alleviating panic in the context of disasters, but instead trigger people's frustration. Some disaster reports have become global focus events for political competition, corporate control, and the expression of different opinions (Cottle, 2009). News organizations are also affected by institutional bias and politics, which may produce distorted descriptions and lead to the continuation of misinformation, stereotypes and misunderstandings (Miller & Goidel, 2009; Berlemann & Thomas, 2019). It is obvious that news sometimes does not focus purely on the disaster itself, but under the combined effect of multiple factors. National power, social problems and media attitudes interfere with readers' judgment. Therefore, it is necessary to compare and analyze disaster coverage in different countries and

provide a more appropriate information representation framework. This study is based on the reality of frequent disasters and the need to give full play to the social service function of the media. It uses multimodal discourse analysis to examine the emergency management role of natural disaster news and provide suggestions for optimizing the defects in the reports.

1.2 Research Purposes and Significance

This study aims to summarize the distribution characteristics of symbolic resources in natural disaster news in China and the UK, providing suggestions for optimizing the quality of reporting. By exploring the synergy mechanism of multimodal elements in the disaster context, the explanatory power of multimodal discourse analysis theory is enriched.

1.2.1 Research purposes

This study starts from the perspectives of the media, management departments, and the general public to provide suggestions for effectively disseminating information and stabilizing social order in the context of disasters.

First, this paper aims to provide more reasonable media reporting discourse strategies to enhance the media's information presentation capabilities and emotional appeasement effects. This requires the researcher to sort out the problems of inappropriate elements and redundancy in media reports, and put forward corresponding suggestions.

Secondly, it is committed to constructing positive and responsible images of social management agencies. The purpose is to enhance the appeal and credibility of them in social security management under the disaster context and strengthen the cohesion of the masses. It introduces their contribution to disaster warning, disaster relief and post-disaster reconstruction, which is conducive to enhancing people's trust and support for them.

Finally, it contributes to arousing public's vigilance against natural disasters, and providing references for improving the public's safety knowledge and rescue capabilities, so as to carry out self-rescue and mutual rescue in disasters. This requires that the distribution of symbolic elements in the news be clearer and more reasonable, making security knowledge attractive and operational.

1.2.2 Research significance

The theoretical significance and practical application values are explained here. It provides feasible suggestions for media to release information, clarify rumors, and reduce negative public opinion in disaster situation.

(1) Theoretical significance

Under the guidance of multimodal discourse analysis theory, this study uses corpus-based

method to analyze natural disaster news, expanding the application of the theory and providing references for the integration of linguistics and communication studies.

It demonstrates the meaning construction mechanism of language and multiple image symbols in news discourse and clarifies the role of multimodal collaboration in highlighting the main information.

(2) Practical application values

This study brings feasible suggestions to the three main bodies of media, social management departments and the general public in dealing with disasters. It can promote them to adopt a collaborative emergency management approach in disaster response and enhance their sense of social responsibility and service.

It helps to understand the tendency and information organization strategies of Chinese and British media in reporting disasters. It encourages the media to narrate from the perspective of accident survivors and spread positive events, which provides references for improving media's awareness of care and the appropriateness of reporting.

1.3 Thesis Structure

This paper is divided into six chapters, and the basic contents involved are as follows:

The first chapter is the introduction, including the research background, research purposes, research significance, and organizational structure of the thesis.

The second chapter is the literature review, elaborating in detail the current status of domestic and foreign research on multimodal discourse analysis and natural disaster news.

The third chapter is the narration to the theoretical basis and research design of this paper. First, the generation, development and application of systemic functional grammar and visual grammar are presented. Then, research methods, research questions and research steps are described. The specific corpus source and annotation process are explained in this chapter in the form of pictures.

The fourth chapter is a comparative analysis of language resources in natural disaster news, which is one of the key contents of this study. The researcher obtains the proportions of the three meta-functions of language resources in China Daily and BBC News. Then transitivity, modality system, theme-rheme structure, etc. are compared with examples to summarize the emergency management effect of language resources.

The fifth chapter explains the proportion of three meanings of image resources and their role in disaster response based on visual grammar. According to the representational meaning, interactive meaning and compositional meaning, typical cases are selected for in-depth analysis. Text analysis helps interpret the role of images in presenting disaster information and rescue

measures, calming people's emotions, and popularizing safety knowledge.

The sixth chapter concludes the whole research by summarizing the interaction mechanism and existing deficiencies of language and images resources. This includes the summary of the main findings, research implications, limitations, and suggestions for future research.

2. Literature Review

The academic community's exploration of the research topic and theoretical foundation provides a general direction for subsequent research. The researcher first introduces the manifestations and characteristics of multimodal discourse, and then discusses the origin, application fields, and research perspectives of multimodal discourse analysis (MDA), which is the key to grasping the entry point of text analysis. Secondly, the causes and classifications of natural disasters, as well as the strategies and impacts of disaster reporting, are explained. It helps the researcher grasp the research hotspots and limitations of this topic and further clarify the collection standards of the corpus. Moreover, the connection points and research focuses of multimodal discourse analysis and natural disaster news become clearer and more focused.

2.1 Multimodal Discourse Analysis (MDA)

Multimodal discourse analysis focuses on the visual, auditory, and tactile interactions between people and external information. The emergence of this theory is closely related to the diversification of information representation methods. It is widely used to analyze pictures and videos of television, movies, advertisements, and speeches. Since its introduction to China, its analysis framework has been continuously enriched, and the research objects have also expanded to daily conversations between individuals or organizations.

2.1.1 Definition of multimodal discourse analysis

“Modality” focuses on the representation form of information. Single text descriptions are rather boring and are gradually unable to meet readers' diverse reading preferences. Technological development has prompted information to be gradually expressed in multiple forms such as text, images, audio, and video. Novel, exaggerated, and interesting symbols cater to the public's aesthetic and cognitive needs. MDA is an emerging paradigm in discourse research that extends the study of language to its combination with other resources such as images, signs, gestures, movements, music, and sounds (O'Halloran, 2011). This study will follow O'Halloran's definition and take into account social-cultural elements when analyzing corpus to ensure that the research content is broad and inclusive.

2.1.2 Research on multimodal discourse analysis abroad

In the 1950s, Halliday establishes systemic functional grammar, which becomes the main theory of multimodal discourse analysis. In 1996, Kress and Van Leeuwen put forward visual

grammar in their book *Reading Images: the Grammar of Visual Design*. They elaborate on the analytical basis of image elements and promote multimodal discourse analysis to a stage of vigorous development. Kress and Van Leeuwen (1996:45) divide the meaning of images into representational meaning, interactive meaning and compositional meaning. Interpersonal communication also has multimodal characteristics, with gestures, facial expressions, etc. reflecting the intentions and characters of the communicators. Norris (2016) analyzes different levels of mediated actions of communicators in family video conferencing to explain multimodal interaction and the mechanism of establishing emotional intimacy. The following is an introduction to the research hotspots of multimodal discourse.

Regarding multimodal critical discourse analysis (MCDA), Machin (2013) believes that the most important point in MCDA is that different types of symbolic resources are expressed through deletion, addition, substitution, etc. Krysanova (2023) proposes that horror in movies is achieved through the coordination of auditory, visual, verbal, and non-verbal modalities such as distorted expressions, screams, trembling, and dim light. Djonov & Zhao (2013:8) think that the combination of critical discourse analysis (CDA) and MDA links multimodal communication, popular culture, and social power, thus generating social critical significance. Keles et al. (2023) code the textbook series “New Headway” and find that women are discriminated against in recreational, domestic, professional, and educational spaces, such as their stereotypical role as caregivers. Lots of research focuses on speeches by politicians and entrepreneurs, analyzing the control and intervention ideas that subtly permeate the speakers’ speeches. Multimodal critical discourse analysis helps researcher discover the power oppression and ideological concepts reflected in phenomena such as gender inequality, violence, and terrifying situations.

Regarding the development from metaphor to multimodal metaphor, Lakoff & Johnson (1980) first propose conceptual metaphor from a cognitive perspective. Forceville combines conceptual metaphor with non-symbolic resources to propose the concept of “multimodal metaphor”, which attracts the attention of the academic community (Koller, 2009). Forceville (2007) elucidates the multimodal meaning of the combination of language, images and non-verbal sounds in commercials and explains the verbalization of multimodal metaphors. Abdel-Raheem (2021) indicates that symbols in political cartoons can present face attacks or face enhancements, while conceptual blending is important for understanding social control phenomena in linguistic or non-linguistic forms. These studies have explored the thinking and cognitive factors in multimodal communication and tried to combine them with metaphor and psychology.

Regarding multimodality and news discourse analysis, Caple & Knox (2015) use systemic functional semiotics to analyze news images, the relationship between text and image, and cross-image sequences. Samuel (2017) analyzes the multimodal discourse characteristics in news

cartoons, pointing out that visual symbols are important for enhancing the comprehensibility of sensitive political issues and highlighting the satirical meaning. Facchinetti (2021) believes that multimodal elements such as pictures, videos, and audio in news enrich information content and promote knowledge dissemination. Bashir & Yasmin (2023) compare the language and image strategies in the epidemic reports of the two countries, revealing their visual features and ideological differences. Multimodal news images bring obvious visual impact to readers, and their meanings are often implicit and diverse. It is necessary to comprehensively consider the combination rules and positions of images in order to explore their deep meaning.

Multimodal information dissemination is closely related to the advancement of visual communication technology, which can highlight specific elements and reveal the designer's attitude. It can be seen that the theory of multimodal discourse analysis starts early and develops rapidly abroad, and has differentiated into multiple branches. In particular, multimodal critical discourse analysis plays an important role in interpreting ideological biases and power relations in cartoons and news. Moreover, the theory is gradually associated with key points such as metaphor and concept blending in cognitive linguistics. Multimodal discourse analysis often uses the computer software Elan to annotate audio and video.

2.1.3 Research on multimodal discourse analysis at home

Research on multimodal discourse analysis starts earlier abroad, while domestic scholars have continuously innovated and expanded the theory. Li (2003) publishes *Social Semiotic Analysis of Multimodal Discourse*, which is one of the earliest semiotic methods for analyzing images in China. Subsequently, many scholars (Zhu, 2007; Zhang, 2009; Feng et al., 2014) enrich and develop multimodal discourse analysis. The key to multimodal discourse analysis is to determine the relationship between different images or various components in the same picture, and the relationship between text and image (Zhu, 2007).

In terms of the construction of multimodal theoretical framework, Zhang (2009) establishes a MDA framework including context of culture, context of situation, semantic level, the formal level etc., dividing the relationship between modalities into complementary and non-complementary relationships. Feng (2014) indicates that MDA gradually integrates disciplines such as linguistics, media studies, and cognitive psychology, developing rapidly with the help of computer simulation and corpus technology. Scholars have established a highly adaptable framework based on the research object and purpose and have focused on the integration of multimodality with other disciplines. This topic has been deeply explored in communication studies, especially in the fields of television and film studies.

In terms of multimodality and corpus research, Gu (2013) proposes that multimodal corpus

refers to the integrated processing of text, audio and image corpus, and users can search in a multimodal way. Zhang et al. (2023) develop a sign language vocabulary feature encoding scheme and build a visual sign language multimodal feature annotation system. Peng et al. (2024) think that building a multimodal traditional Chinese medicine (TCM) corpus can not only collect TCM terminology translations, pictures, videos, etc., but also fully demonstrate the unique charm of TCM. Multimodal corpora have strong practical application value because they are closely related to social issues and cultural protection. With the help of computer tools, the TCM corpus retains specific terms in a certain era, cultural background, and region.

In terms of multimodal news analysis, Wang (2019) analyzes the discourse environment, discourse subject and constituent elements of government new media photos based on visual grammar. Liu (2015) highlights that applying multimodal discourse analysis to news pictures can interpret the process by which news pictures influence readers' perception through preset reporting perspectives. Pan (2019) interprets the differences in narrative mode and discourse strategy between Chinese and American military news from the perspective of multimodal cognitive critical discourse. Domestic multimodal research focuses more on political, military, and entertainment news, believing that differences in multimodal forms can affect the perception of the audience.

In summary, most studies have noted that language information and symbolic resources can highlight key information in news through effective arrangement, and establish psychological connections with readers through modality, voice and gaze. The widespread use of multimodal resources reflects an audience-oriented trend. Research related to this theory mainly concentrates on the field of linguistics, lacking interdisciplinary awareness. Multimodal discourse analysis is widely used in oral and written text analysis, such as daily conversation, advertisements, documentaries, etc. It is increasingly focusing on the language of special groups, such as the elderly, aphasic patients, sign language users, and the construction of Chinese traditional cultural corpus. The theory has also been continuously developed, but it still takes systemic functional grammar and visual grammar as the main framework. As far as multimodality and news discourse analysis are concerned, most studies select a specific news text, with a small number of samples and lack of strong data support. Therefore, this paper interprets the emergency management functions of news through multimodal discourse analysis of large-scale corpus.

2.2 Natural Disaster Reporting

This chapter mainly introduces the definition, classification and characteristics of natural disasters. Then, the current research status of natural disasters at home and abroad is introduced in a hierarchical manner. In particular, the relationship between natural disasters, social management

departments and news agencies is discussed in detail, which clarifies the importance of natural disaster news and the limitations of current research.

2.2.1 Definition of natural disasters

Natural disasters originate from natural processes, which can affect community safety and have negative consequences, undermine social order and cause human and material losses (Chaudhary & Piracha, 2021). Domestic and foreign scholars have different definitions of natural disasters, but all pay attention to the strong destructiveness and suddenness of them. Natural disasters are events that cause casualties and property losses to human society due to natural events or forces, including 13 categories such as floods, droughts, earthquakes, fires, typhoons, and sandstorms (Huang, 2009). Mata-Lima et al. (2013) believe that natural disasters are caused by hydrometeorological, climatic, geophysical and biological phenomena, which have an adverse impact on the environment and buildings in the affected areas. Natural disasters are inseparable from emergency management because it is an important means to reduce losses and protect the vital interests of the people.

2.2.2 Research on natural disaster news abroad

It is necessary to understand the current status of research on natural disaster news, because natural disaster prevention and response is an important topic in the context of sustainable development. The writer has read extensively on emergency management and news reporting in academic platforms, and the current hot topics are summarized into the following three aspects. They cover the content, strategy, role and research of natural disaster reports in the field of linguistics. At the same time, the shortcomings in disaster reports have also received attention. This helps the researcher understand cutting-edge research on related topics and find potential directions worth exploring from a linguistic perspective.

In terms of the emergency management role of the media, the media is a necessary channel for people to obtain information and express their demands in the context of disasters. Studies have shown that the media has a strong positive and certain negative role in emergency management. Luna & Pennock (2018) raise that news reports enable audiences to obtain disaster information in a timely manner, and also enable decision makers to monitor public activities and promote coordination among all parties. Olanrewaju & Ahmad (2018) find that tweets posted on social media are conducive to providing emotional support, contacting disaster victims, coordinating material distribution, and enhancing community resilience. As far as negative effects are concerned, Hiltz et al. (2012) deduce that social media reports on disasters have problems such as false information, information overload, exposure of privacy, and violation of ethical principles.

Disaster reports have a dual nature: on the one hand, it accelerates the dissemination of information and the cooperation of relevant responsible parties; on the other hand, it sometimes hides the truth and hurts the emotions and privacy of the victims.

In terms of the content and strategy of disaster reporting, current disaster reporting shows the characteristics of gradually enriching content, channels and forms. Liu & Stevenson (2013) conduct a cross-cultural study on earthquake reports in Australia and China and find that their reporting tendencies show differences due to cultural background. Scott & Errett (2018) hold that Facebook posts during flood disasters provide information such as material supply, evacuation routes and emergency measures. Disaster reporting provides people with necessary emergency measures and escape routes, becoming a significant tool to help social rescue and mass self-rescue in disasters. In different cultural backgrounds, reports on similar disasters have their own tendencies and cultural characteristics.

In terms of disaster reporting and credibility building, current research mainly involves government credibility and media credibility. The former refers to the influence and appeal of the government in public affairs; the latter refers to people's trust in media reports. Endsley et al. (2014) conclude that people believe that disaster information from different sources has different credibility, and traditional media such as television news and print news are more credible. Wajid & Taimur (2016) confirm that the factor that most easily affects the credibility of social media is the disaster and relief news it publishes. Research shows that people are more inclined to trust information provided by traditional media. The authenticity and timeliness of disaster reports have an impact on the reliability of the media and the government.

In the field of linguistics, Puspita & Pranoto (2021) find that judgment is the most common attitudinal resource in Japanese news, followed by appreciation and emotion, which reveals the newspaper's attitude and evaluation of the event. Öhman et al. (2016) conduct a critical discourse analysis of fire-related news, expounding the struggle between risk discourse, discourse legitimacy, and crisis communication in power discourse. Nazaruddin (2017) utilizes visual discourse analysis to examine news on volcanic eruptions, focusing on how images convey the situation of volcanic eruptions, evacuation processes, and evacuation centers. Most of the studies are conducted under the framework of functional grammar, combining corpus technology with appraisal theory and critical discourse analysis to enhance the persuasiveness of the analysis results.

In summary, media organizations attract the attention of people from all walks of life by publishing news in the context of disasters. Disaster reporting strategies, content, emotions and the subject's views on the event have an obvious impact on information dissemination and spiritual comfort. Foreign scholars have systematically examined issues such as the application of technology and equipment, journalistic ethics, and crisis evolution in the context of disasters. From

a linguistic perspective, corpus-based research has enabled news discourse to be explained more comprehensively. These studies explore in depth the social and cultural factors, safety concepts and care awareness reflected in disaster news. However, the overall analysis of elements in news images is insufficient, reflecting that research has not paid enough attention to it.

2.2.3 Research on natural disaster news at home

In China, the types and frequency of natural disasters have obvious regional characteristics. Wei & Wang (2009) divide common natural disasters into geological disasters, meteorological disasters, fires, marine disasters, biological disasters, etc. Disaster prevention and response have attracted the attention of many scholars in the field of management. Domestic research focuses on public opinion guidance, emotional encouragement, and the display of positive government images under the background of disasters. Considering that this study focuses on media reporting strategies, the image of individuals or institutions, reporting and disaster response, the frontier research will be introduced from the following aspects.

In terms of disaster reporting and media communication, Zhong (2008) emphasizes that information release and public opinion guidance are very important for reducing hazards, enhancing information transparency, and grasping the initiative of public opinion. As an important part of media communication, emergency reporting is very critical for transmitting information, maintaining public safety, assisting rescue, and spreading positive social energy (Huang, 2023). Wang (2025) holds that the emergency science works released by the media are not standardized, and have problems such as content homogeneity and incorrect timing, which can lead to biased information interpretation. Scholars generally believe that news media have significant responsibilities in various links of emergency management, such as risk warning before disasters, information dissemination when disasters occur, and emotional encouragement in post-disaster recovery. The positive role played by the media in emergency management is far greater than the negative influence.

In terms of emotional guidance in disaster reporting, Wen et al. (2023) confirm that inaccurate information is prone to generate negative emotions such as panic, anxiety, and anger, which hinders the governance of emergencies. Guo & Yang (2016) propose that disaster news is not necessarily “neutral” because media use language to inject their thoughts or ideologies into the text, affecting readers’ views on the event. Yan & Shen (2023) say that disaster news must take into account the humanistic care in report and adopt a narrative strategy that integrates objectivity and emotion. Li & Fu (2022) apply the frame theory to analyze news coverage and draw that local media pay more attention to emotional motivation and are accustomed to reproducing heartwarming scenes at the disaster site. Language can reflect the media’s reporting tendency and

attitude towards events. The quality of news is crucial for establishing the responsible image of government departments and enhancing the credibility of the media.

In terms of disaster reporting and institutional image building, in sudden natural disasters, due to the lack of government credibility and the absence of official media, the public is easily misled, which promotes the eruption of public opinion and damages the image of the government (Shan & Zeng, 2019). Xu & Xu (2020) suppose that timely and transparent information disclosure by the government is conducive to improving public satisfaction and government credibility. Zhao (2014) believes that the timeliness, authenticity, and completeness of information in disaster reports are the “touchstone” for testing the credibility of the media. However, some news reports are not objective enough, and their narratives have strong personal subjective tendencies. Inappropriate news can easily trigger rumors, leading to the outbreak of negative emotions, which is not conducive to risk control.

In terms of emergency discourse, the cultivation of emergency language service talents and emergency language ability has received wide attention. Dong (2021) defines emergency language as a language that provides convenient services for prevention, control, monitoring, and rescue activities in sudden natural disasters and public events. Emergency language ability plays a prominent role in implementing language assistance, providing information services, and responding to public opinion and discourse conflicts (Wang, 2020). Zheng & Yuan (2024) analyze the vocabulary, semantics, and grammatical characteristics of foreign-related police emergency rescue, and explore how to enhance their emergency rescue language ability. Emergency language service talents refer to people who can use relevant knowledge, skills, and experience to engage in emergency language services (Zheng & Xu 2020). Language is closely related to emergency management, and dialects, minority languages, and intelligent language software are important for carrying out medical treatment and rescue in emergency situations.

In summary, domestic and foreign scholars have made rich research achievements on disaster news, and these studies are concentrated in the fields of management and communication. Through the integration and classification of current research, it is concluded that emergency news research in the field of linguistics has not received enough attention at present, especially the discourse strategy, narrative framework, and symbol combinations of such news deserve further discussion. Disaster response is a major challenge faced by management departments and the media, and related research shows a cross-cutting trend in engineering and social sciences.

3. Theoretical Basis and Research Design

The previous chapter describes the relevant content of MDA, which helps the researcher understand the branches and applications of the analysis framework and grasp the entry point of subsequent text analysis. Secondly, the detailed analysis of disaster and news related research illustrates the significance of this topic. Reasonable theoretical support and research design are necessary conditions for obtaining meaningful research results. In addition to introducing the theoretical basis and research methods, this chapter also shows corpus collection, annotation, research questions, etc.

3.1 Theoretical Basis

Theory provides a basis for text analysis, making the interpretation of language and symbols regular and orderly. Halliday (1973:42) proposes the ideational function, interpersonal function and textual function of language in systemic functional linguistics (SFL). Kress & Van Leeuwen (1996:22) propose visual grammar (VG), dividing image meaning into representational meaning, interactive meaning and compositional meaning, which provides a basis for analyzing the role of image elements in the context of disasters. These two theories are closely related and constitute the multimodal discourse analysis framework, which provides a basis for explaining the responsibility of language and images in disaster news.

3.1.1 Systemic functional grammar

Systemic functional grammar (SFL) is created by Halliday inheriting the views of Firth of the London School (Li, 2008). As its name suggests, this theory emphasizes the hierarchical system and functions of language. Systemic functional grammar has gone through three stages of continuous development. The first stage is “order and category grammar” (Huang, 2000; Xin & Huang, 2011), including three orders and four categories. The second stage refers to systemic grammar, in which Halliday sorts out the relationship between surface structure and deep structure (Huang, 2000). Halliday (1973: 49) mentions that language is the symbolic resource for creating meaning, and advocates that language is a series of possibilities and an open system that individuals can choose. The language choice of news writers is flexible and active, which is influenced by factors such as the seriousness of the disaster context and the diversity of the public’s cultural background. As for systemic grammar, Miao (1998) assumes that the language system is manifested as a selection relationship, the choice is primary, the structure is the result of various choices, and the meaning is the choice in a certain context. The third stage is the functional grammar stage, where Halliday proposed the meta-functions of language, which has a profound

impact on language research.

Ideational function refers to the fact that communicators can use language to express the consciousness of the inner world and the experience of the external world and the basic logical relationship therein (Halliday, 1973: 66). Ideational function is expressed through the transitivity, including material process, mental process, relational process, behavioral process, verbal process and existential process (Halliday, 2004: 179). The material process describes the situation in which an “actor” performs actions such as “hit, move, kick”; the verbal process refers to a “sayer” expressing information through speaking to the “receiver”; relational process describes certain characteristics or identities of things (ibid:227); the mental process expresses the psychological state of a person, such as perception, desire, cognition; the behavioral process refers to physiological activities such as breathing, crying, and dreaming (ibid: 248); the existential process indicates that a certain event exists in the psychological or real world, with a relatively obvious “existence” sign. The function of the environment in the transitivity is to supplement the information expressed by the process and the participants, which is equivalent to the adverbial component in syntax (Sun & Guo, 2022).

Interpersonal function refers to the speaker’s use of language as a means of intrusion in speech events: expressing his comments, attitudes and evaluations (Halliday, 1973:106). It expresses the communicator’s status, identity, attitude, etc. through the modality and mood systems (Halliday, ibid:41). For example, when a speaker seeks help from an elder or a higher-ranking employee in the company, he or she tends to use a euphemistic, intimate tone and polite words. Modal words have differences in polarity, and Halliday(2004:146) believes that polarity is the possibility of choosing between the two levels of “yes” and “no”. Halliday (2004:350) divides modal verbs into high-value, medium-value and low-value modal words. The modal verbs contained in different levels of modality and the attitudes they display will be explained in Chapter 4.

Textual function refers to a speaker organizing independent and separate words, phrases, clauses, etc. to form the discourse that conveys meaning in the context. Semantically coherent and logical news with prominent core content not only reduces obstacles for readers to understand, but also reflects the excellent language organization skills of the news writer. Textual function is realized through thematic structure, information structure and cohesion (ibid:82). Halliday’s classification of thematic types includes singular/plural themes, marked/unmarked themes, and special themes including thematic correspondence and predicate themes (Fu, 2009). Huang (2001) explains the structure of “overlapping themes”, namely wh-question components, wh-relative components, and anaphoric components.

Systemic functional grammar is widely used in news discourse analysis. Mineshima (2009) studies the interpersonal function of political news, explaining how news provides hidden support

for the writer's position and attracts readers' recognition through lexical grammar and text organization. Potter (2016) uses Halliday's theme-rheme to analyze the information structure in news related to the death of Lebanese national leaders, proving the effectiveness of subject-rheme in revealing potential news ideology. Ansary & Babaii (2005) conduct a genre analysis of *Washington Times* news, emphasizing that systemic functional grammar links the contextual dimension of register with the semantic and grammatical organization of the language. Fatima et al. (2018) explore the transitivity of news on attacks in Pakistan and finds that material process dominates in Dawn, while verbal process accounts for the highest proportion in other news. Chen & Wei (2024) draw on systemic functional grammar to construct a multimodal harmony analysis framework to analyze peace, restorative narratives, and expectations in earthquake news. This theory has currently spawned new branches, such as appraisal theory and multimodal social semiotics. It has become the theoretical basis for analyzing Chinese and British news discourse. This study will refer to the analytic paradigm in the above content and analyze the construction of disaster news discourse with the help of transitivity, modality, theme-rheme structure, etc.

3.1.2 Visual grammar

Visual texts often appear in the form of multimodal compositions, which combine written language, design elements and visual images (Serafini, 2014:2). Communication is a process of transmitting, receiving and deciphering certain forms of signals that interact with each other by sense of smell, touch, hearing, vision to promote the understanding of meaning (Pettersson, 1993). In fact, contemporary media information is characterized by a high degree of semiotic complexity and combines various modes arranged in different ways. Holsanova (2012) believes that the perception of vision is determined by bottom-up factors (such as form, color, brightness, contrast) and top-down factors (such as interests, attitudes, motivations, prior knowledge, cultural differences). In disaster news, there are a large number of meaningful images composed of elements such as points, lines, and surfaces, which stimulate the imagination in the readers' minds and cause their emotional responses.

Kress (2009) argues that modality is a socially and culturally shaped symbolic resource used to express meaning. Images, text, layout, music, gestures, speeches, dynamic images, and audio are all commonly used modalities in communication. One of the earliest foreign scholars in multimodal discourse analysis is R. Barthes. In his paper *Rhetoric of the Image*, he explores the interaction between images and languages in meaning expression (Li, 2013). According to the theory of visual grammar, representational meaning refers to the fact that image symbols can reproduce what happens in the subjective world and the objective world. Depending on whether there is a vector in the image, it can be divided into narrative representation and conceptual

representation (Kress & Van Leeuwen, 1996:45), which corresponds to the ideational function in systemic functional grammar. Vectors are usually expressed through limbs, eye contact, tools used in the action, and body movements (ibid: 59). There are vectors in narrative representation, which objectively presents what happens in the real world and reflects a dynamic process. However, there are no vectors in conceptual representation, which is usually stable. Conceptual representation represents the relationship between participants, concepts or objects from the perspective of spatial organization, and is used to classify people or things (Serafini, 2014:62). Interactive meaning involves the emotional response generated by the interaction between the viewer and the image elements, which is expressed through three elements of distance, contact and perspective. Compositional meaning refers to the ability of the arrangement and combination of symbols to disseminate information to varying degrees, mainly in terms of information value, framing, and color salience. The misalignment, surrounding, inlaying and overlapping relationships of images affect the prominence of information.

Liu (2017) points out that visual rhetoric emphasizes the identification and analysis of “visual form” and excavates the implicit meaning hidden in the “rhetorical structure” of visual texts. Friedman & Ron (2017) apply visual grammar to analyze U.S. presidential election advertisements, illustrating the role of graphic structure and information distribution in attracting voters’ attention. Barcelos & de Azevedo (2021) adopt animated films to develop curriculum to enhance students’ visual literacy and critical thinking skills. Qadha & Al-Wasy (2024) find through experimental research that the group using visual grammar tools achieves significant improvements in learning participial adjectives.

In summary, discourse is certainly the main form of information representation, but images, music and colors are more likely to arouse people’s interest and curiosity. The combination of images and texts reflects the importance of information and triggers people’s associations and imagination. The three meanings of images correspond to the three meta-functions of language, forming a clear multimodal discourse analysis framework. The meta-functions of language and visual grammar can be used to analyze text and image elements in the discourse respectively. This framework has a relatively complete system and strong applicability. The systems of the two theories have been continuously enriched, and many scholars have combined them to analyze the meaning of symbol design in movies, speeches and advertisements. The theoretical framework is highly adaptable to the analysis of news texts, providing a basis for revealing the basic meaning and implicit meaning of disaster news and the attitudes of the speakers.

This paper will sort out the meaning and symbolic expression of the elements involved in the theory. Then, with the help of corpus tools, the language and images that meet the properties of the element are annotated and the connection between the data is explained. Finally, typical cases

are selected through text reading and the roles and responsibilities of language and images are respectively demonstrated under the guidance of the theoretical framework.

3.2 Research Design

This chapter introduces the concepts, operation steps and functions of research synthesis method, corpus-based method and discourse analysis method. The research design is based on intensive literature reading and large-scale corpus. This helps the researcher identify problems and adopt appropriate paradigms to explore in steps, thereby finding the similarities and differences in the construction of disasters by Chinese and British media. In terms of corpus collection, this study selects various types of natural disasters as search terms on two platforms to ensure the diversity of the corpus. The following are two research questions:

- (1) What are the proportions of the three meta-functions of language resources and their emergency management functions in the two corpora?
- (2) What are the proportions of the three meanings of image resources and their emergency management functions in the two corpora?

3.2.1 Research methods

Research synthesis method is an empirical method in which researchers extensively collect and read literature. They need to systematically analyze previous research results on related topics to generate comprehensive understanding. According to the research questions, the writer uses CNKI, Google Scholar, Web of Science and other platforms to conduct literature search and intensive reading. It helps her understand the current status and shortcomings of research on multimodal discourse analysis and disaster news, and clarify the meaning and framework levels of elements.

Corpus-based research is a top-down analysis method that can be summarized into seven steps, namely, determining analysis framework, segmenting discourse units, identifying functions of independent units, examining vocabulary and grammatical features, describing language features of each functional category, performing sequence analysis, and describing the general pattern of discourse (Upton & Cohen, 2009). The Uam Corpus Tool 3.3x and the Uam Image Tool 2.1 are used to establish the annotation hierarchies, annotate the corresponding components and count the results, which lays the foundation for further analysis.

Discourse combines dimensions such as form or expression (sound, vision, words, phrases, etc.), meaning and action (Van Dijk, 2011). Through discourse analysis, specific ways for news to organize language, construct attitudes, and guide readers to form opinions can be obtained (Ding, 2019). The writer will select typical cases through text reading and analyze the vocabulary

selection, sentence structure, tense and person of Chinese and English disaster news.

3.2.2 Research procedure

According to the research method, this study determines the research steps, including the preliminary collection of China Daily and BBC News corpus, text and image annotation, data statistics and interpretation, case analysis, etc. The research process ensures that the analysis of news texts is from macro to micro, that is, from the overall data description to the interpretation of specific symbolic meanings.

(1) Corpus collection

Ensuring that the corpora are sourced from a comparable level of media is an important step. China Daily and BBC News are very influential in the world, with a large number of readers and wide circulation. BBC News is one of the largest news media organizations and a representative of Western mainstream media (Zhu & Huang, 2017). China Daily is the only Chinese newspaper that has effectively entered the international mainstream society and has the highest reprint rate in foreign media (<https://baike.baidu.com>). The news reports they publish have a certain authenticity and representativeness, serving as a window for people to understand important global information. Therefore, comparing the disaster news published on these two platforms can help reveal common problems and propose solutions that are easy for the news industry to adopt and learn from.

The writer uses earthquakes, tsunamis, landslide, floods, hurricanes, wildfire, volcanic eruptions, etc. as search terms and the search period is set from 2019 to 2024. News of natural disasters from China Daily (<https://www.ChinaDaily.com>) and BBC News (<https://www.bbc.com>) are downloaded. News with about 500 words will be retained, while news with too many or too few words will be excluded. The downloaded news is saved in TXT format so that it can be successfully recognized by the corpus tool. Then the blank lines and garbled characters are deleted, and the spelling errors caused by word extraction are proofread.

This study has obtained 60 natural disaster news respectively from China Daily and BBC News as research corpus, including 64,221 words and 160 pictures. The explanatory text of the news pictures will also be retained to provide necessary event, time and location information for further analysis. The types and numbers of disasters in the China Daily and BBC News corpus are shown in Table 3.1. The bar chart shows that floods, wildfires and storms are the most reported natural disasters by China Daily and BBC News, with 36 and 35 reports respectively. The least reported natural disaster by China Daily is drought, while the least reported disaster by BBC News is hurricanes.

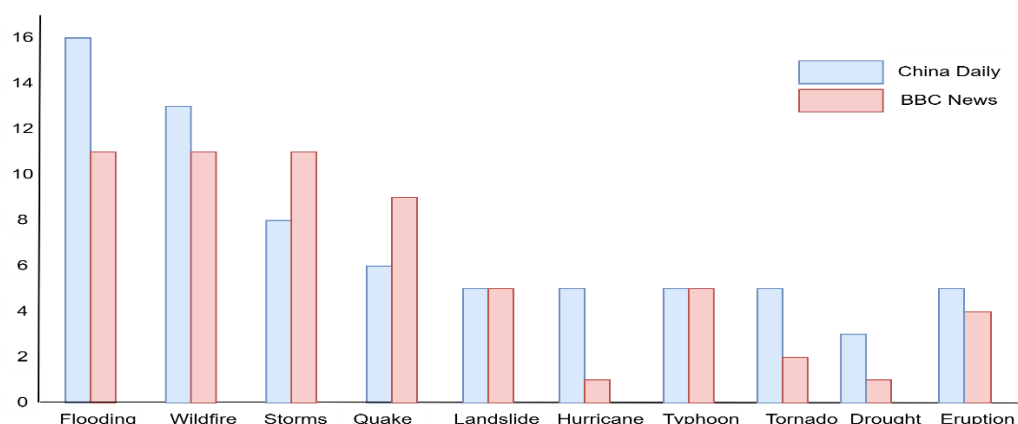


Figure 3.1 The number and types of natural disasters in the collected corpora

(2) Corpus annotation

The corpus tools used are Uam Corpus Tool 3.3x, Uam Image Tool 2.1, and Antconc 3.5.9, which have clear interfaces, convenient operation, and can reduce errors caused by the researcher's personal subjective judgment. The writer first adopts the systemic functional grammar annotation layers in Uam Corpus Tool to automatically annotate the text and then manually proofreads it (As shown in Figure 3.2). This greatly improves the quality of the annotation results. Researcher can click the "search" function in this software to locate the specific location of the annotated sentence and view relevant elements. At the same time, researcher can use this function to view the pretext and post text of the sentence to clarify the context and semantic relationship.

Then, based on visual grammar, the image annotation software Uam Image Tool is applied to establish the annotation levels of representational meaning, interactive meaning and compositional meaning. The writer first clicks "extend corpus" to upload pictures, then clicks "add layer", "annotate parts of image" and "edit" to create and adjust the annotation elements as needed. Next, according to the judgment of specific elements, the writer selects the corresponding image, and then clicks the theoretical concepts at the bottom of the interface for local labeling, and finally clicks on "search" and "statistics" to view the marked areas and data. (As shown in Figure 3.3).

The researcher imports a stop word list into Antconc to filter out the function words in the two corpora, reducing their interference with the research results. Then, she imports the BNC (British National Corpus) as a reference corpus, counts the keywords of the two corpora respectively, and sorts them according to their keyness (As shown in Figure 3.4). Next, she enters "can, should, may" and so on one by one in the "search term column" of Antconc to retrieve modal verbs in the two corpora (As shown in Figure 3.5). Antconc's concordance function can count the frequency of the search terms. Finally, she checks the annotations, corrects the errors, exports the data, and fills in the charts. The overall analysis framework is shown in Figure 3.6.

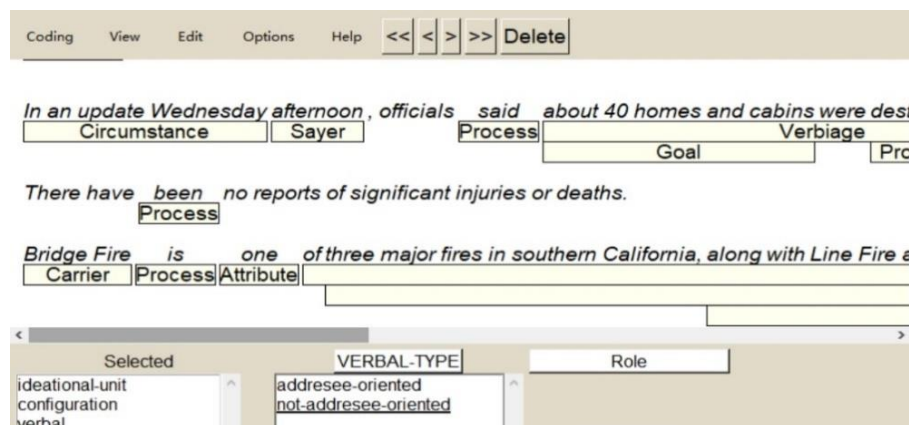


Figure 3.2 Operation of Uam Corpus Tool 3.3x



Figure 3.3 Operation of Uam Image Tool 2.1

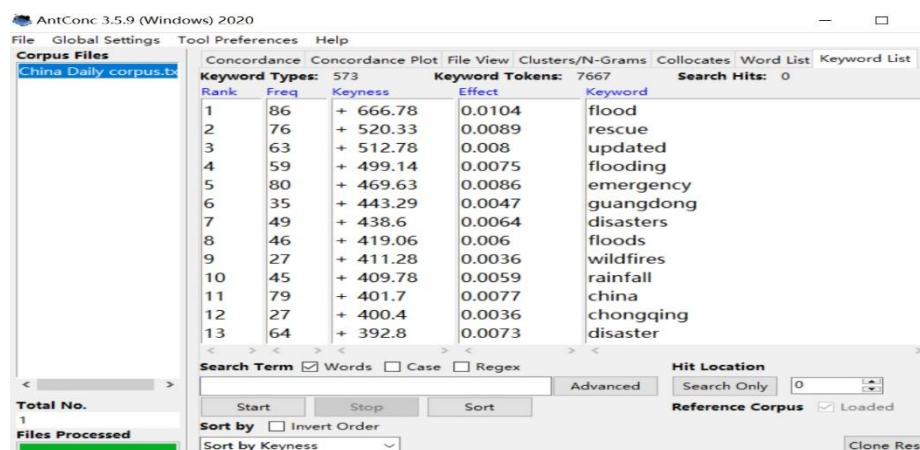


Figure 3.4 Keyword statistics operation of Antconc 3.5.9

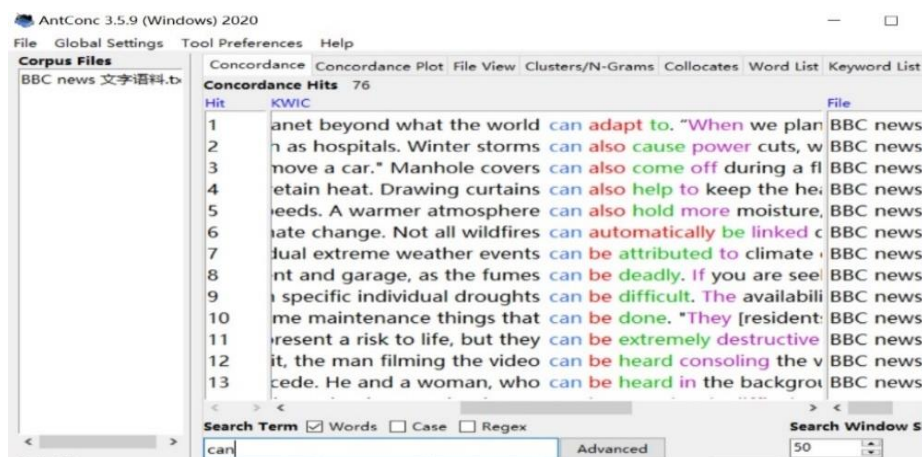


Figure 3.5 Modal word statistics operation of Antconc 3.5.9

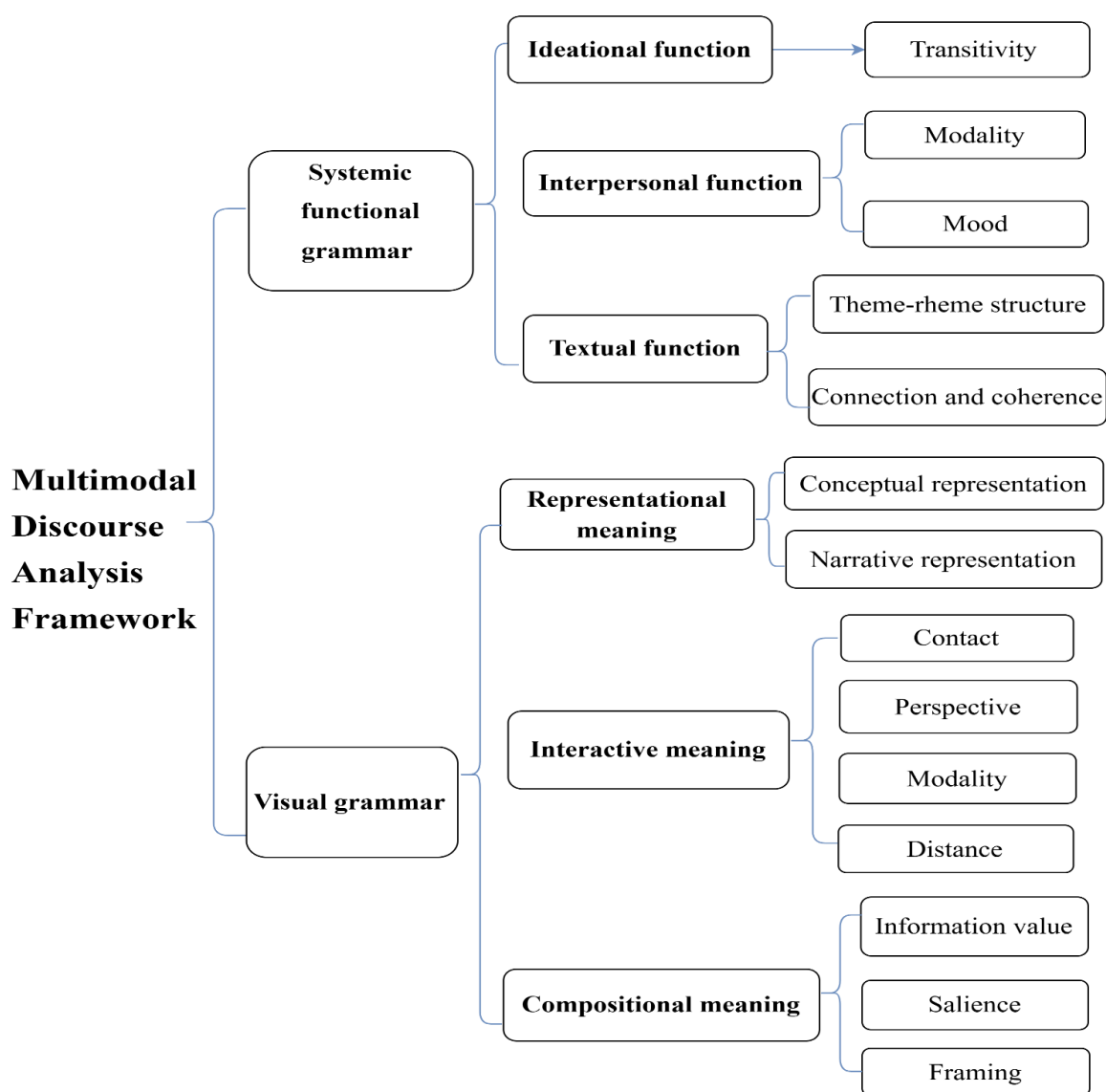


Figure 3.6 Analysis framework of natural disaster news

In the above content, the researcher sorts out the application and role of the three research methods, the steps involved in the research procedures and their corresponding purposes. In particular, the types and numbers of natural disasters involved in the two corpora and the operation of the corpus tools are explained.

In the following chapters, the proportion of discourse and image symbols in natural disaster news and their role will be explained. Typical examples of discourse and images, along with their corresponding natural disasters, are recorded for subsequent analysis. Its purpose is to form an objective and comprehensive understanding of disaster discourse and propose a more appropriate reporting paradigm. The quantitative research helps the writer conduct a statistical and descriptive analysis of linguistic and non-linguistic symbols, which is somewhat objective and representative.

4. Comparative Analysis of Natural Disaster News Language

Based on systemic functional grammar, the writer has annotated the ideational function, interpersonal function and textual function of discourse in natural disaster news and obtained their proportions. Transitivity, modality, mood, and various theme-rheme structure will be combined with examples to analyze the emergency management role of discourse. Through in-depth analysis, the lexical and grammatical features of disaster news and the characteristics of sentence structure can be understood. The reproduction of the disaster process in the news and the speaker's inference of the cause and impact of the disaster are observed in this study. More importantly, by analyzing the verbal process of the disaster-stricken people, their most real feelings about the destructive power and impact of the disaster, as well as the demands conveyed in their words, are reflected.

4.1 Ideational Function of News Discourse

The ideational function is expressed through the six processes of transitivity, which comprehensively describe event-related actions and psychological experiences. Transitivity divides people's discourse expressions of what they see, hear, and do in real world into "processes" that can be analyzed, clarifying the relevant participants and environment components of each process (Hu et al., 2005; Li, 2016:191). The concepts of these processes are introduced previously and Table 4.1 shows the frequency and proportion of the six processes of language resources in China Daily and BBC News.

Table 4.1. Frequency and proportion of processes based on transitivity

Process	China Daily		BBC News	
	Frequency (times)	Proportion (%)	Frequency (times)	Proportion (%)
Material process	2172	71.77	2463	65.02
Verbal process	420	13.88	507	13.38
Relational process	267	8.82	505	13.33
Mental process	120	3.97	218	5.76
Existential process	22	0.73	65	1.72
Behavioral process	25	0.83	30	0.79

As can be seen from Table 4.1, the proportions of the six processes in China Daily and BBC News are very similar. In the two corpora, material process accounts for the highest proportion, 71.77% and 65.02% respectively. The proportions of verbal process are roughly the same, 13.88% and 13.38% respectively. The proportion of relational process ranks third, and its proportion in BBC News is 4.51% higher than that in China Daily. Natural disasters often have adverse effects on various living and non-living entities, and news needs to narrate these processes to let people

understand the situation. Relevant departments generally take various measures to ensure rescue, and the reports of these work practices can show the execution of management departments. Therefore, the material process accounts for the highest proportion. Disaster accidents are related to the vital interests of social members, so they will cause heated discussions, among which experts, the public, and officials exchange opinions to convey or obtain information, which leads to a high proportion of verbal processes. Only by identifying and classifying the causes, nature, features, and impact range of natural disasters can professionals develop useful response plans. These aspects need to be explained clearly in the relational process, so this process accounts for a relatively high proportion. The emergency management effects and linguistic features of the first three processes will be mainly explained later.

The proportions of mental process are fourth, 3.97% and 5.76% respectively. In China Daily, the proportion of behavioral process (0.83%) is slightly higher than that of existential process (0.73%), while the proportion of behavioral process (0.79%) in BBC News is slightly lower than that of existential process (1.72%). The latter three processes account for a very low proportion, which is related to the purpose and perspective of news. The purpose of disaster news is to accurately and quickly disseminate true information, avoid excessive emotional exaggeration, and mainly use an objective third-person perspective. However, the latter three processes are mostly from the first person, describing inner emotions and physiological manifestations of people. It is inappropriate to over-describe these contents in the urgent disaster context, which damages the objectivity and conciseness of the news.

Table 4.2 Top 20 keywords in China Daily and BBC News				
Rank	China Daily		BBC News	
	Keyness	Keyword	Keyness	Keyword
1	666.78	flood	942.33	flood
2	520.33	rescue	477.26	landslides
3	512.78	updated	473.35	storm
4	469.63	emergency	439.75	rescue
5	443.29	guangdong	410.25	wildfires
6	438.6	disasters	400.58	people
7	411.28	wildfires	388.49	homes
8	409.78	rainfall	375.39	rainfall
9	401.70	China	361.64	climate
10	400.40	chongqing	286.58	residents
11	392.80	disaster	271.21	typhoon
12	363.80	province	258.11	earthquake
13	347.52	xinhua	239.11	firefighters
14	324.66	typhoon	220.51	hit
15	310.16	earthquake	218.44	weather

16	292.43	hunan	214.69	disaster
17	287.19	heavy	205.13	region
18	245.59	residents	203.08	agency
19	234.21	sichuan	201.26	damage
20	233.53	people	181.11	officials

Keywords reflect the theme of the text, which can better illustrate the lexical characteristics of Chinese and English news. Table 4.2 shows the top twenty keywords in Chinese and English news reports, most of which are related to disaster categories. Disaster keywords include flood, landslides, storm, typhoon. Obviously, the news focuses on the adverse effects of various disasters on residents and regions and the rescue operations carried out by various organizations. Words like “threaten, heavy, hit, damage, residents and people” reflect the strong destructive power of disasters on human beings. There are more location keywords in Chinese news, such as guangdong, chongqing, sichuan, hunan, province. Rescue keywords in British news include rescue, firefighters, agency, officials. These keywords basically reflect the focus of news reports.

4.1.1 Material process

In the context of disasters, it is necessary to identify the subjects and behaviors of relevant events in order to conduct accountability, assign tasks, and reflect on experience, and thus apply more targeted and guided response strategies. The material process objectively describes the process by which the subject of action implements the behavior, including elements such as the participant “actor”, “process” and “goal”. The actor who performs the behavior is called the “behavior subject”, which is represented by the name of the disaster, personal pronouns, official agencies, rescue workers, the number of casualties, etc. in natural disaster news. The “goal” is the object affected by the behavior of the subject, represented by the disaster results, casualties, rescue supplies, infrastructure, equipment, tools, etc. In natural disaster news, material process is mainly used for disaster warning, explanation of the impact of disasters, rescue operations and the living conditions of disaster victims.

When describing the impact of disasters, the process is mainly realized through verbs such as destroy, break, flood, cut, kill, confirm, cause, crush, bury, trigger, etc. When describing rescue efforts, common words that express actions in the corpora include “provide, offer, protect, restore, help, move, allocate, check”, etc. The actions of social members in disaster situations have been sorted out, mainly including “running, hitting, waving, grabbing, pulling, crawling, tearing, etc.”, which reflects the most common reactions of people in emergency situations. The researcher selects three typical cases from China Daily and BBC News respectively, and analyzes the actors, behaviors, goals and results of the rescue in combination with the situational context. The italicized

parts in the example are the main components of the analysis, and they usually correspond to the elements of the process.

Examples:

1. *I was pounding against* the doors with my hands, and *shouted* at the top of my voice, my throat was almost broken, Xiang *recalled*. (China Daily)

2. *A tornado* near the town of Holdenville *killed* two people and *damaged or destroyed* more than a dozen homes, according to the Hughes County Emergency Medical Service. (China Daily)

3. *We have prepared* bread, milk, bottled water and daily necessities and *organized* 42 volunteers to provide services for those in need. (China Daily)

4. *Aid agencies have been working to get relief supplies* to thousands of people in south-east Afghanistan, struck by an earthquake in late June. (BBC News)

5. *Rescuers had constructed* a temporary pedestrian bridge and *used* ropes to rappel across the river to reach stranded people there and move the injured to hospitals. (BBC News)

6. *The storm has damaged* bridges, *torn* roofs *off* buildings, *damaged* factories and *triggered* widespread flooding and landslides, leaving 64 people still missing. (BBC News)

Analysis: Example 1 presents the experience of a community volunteer when a disaster occurs. The actor of the material process is “I”, referring to “a local volunteer” because he is the one who initiates a series of rescue actions. The words that represent the material process are “was pounding, shouted, and recalled”, showing the urgency of the situation. Each verb that represents the material process has a corresponding “goal”, namely, the recipient of the action. The goal of the action “was pounding” is “doors”, and its purpose is to wake up the villagers and arouse their attention and vigilance. The adverbial phrase “with my hands” further highlights the volunteers’ anxious mood to save people at that time. The goal of the action “recalled” is the critical situation when the flood occurs. The action “shouted at the top of my voice” is directed at the community residents, indicating that the volunteer did his best to call the villagers to strive for time for them to evacuate and escape. These series of verbs show the responsible image of community volunteers who disregard their personal safety and try their best to protect the safety of community citizens in disaster.

Example 2 uses a series of verbs to intuitively show us the serious harm caused by the disaster. In this case, the “actor” of the process is an inanimate subject “tornado”. The verb “killed” in the process corresponds to the goal “two people”. The actions “damaged and destroyed” directly point to the goal “homes”. These verbs with negative sense indicate that the tornado seriously threatens people’s lives and property safety. At the same time, example 2 also uses an evidential marker, that is “according to”, indicating the source of information. It introduces an agency “the Hughes County Emergency Medical Service”, which shows that the information comes from an official

institution and has strong reliability. Research shows that various forms of evidential markers such as “it is reported that”, “according to”, “measurement results show that”, and “as stated by” are widely used in news. Many of the reports cite quotes from administration, sector officials, authoritative documents, and eyewitness accounts because they are more reliable and professional.

In example 3, the “actor” of the process seems to be the plural first-person pronoun “We”, which refers to a rescue team. But its narrator is the person in charge of the temporary resettlement site of Pingwu Vocational School, who does not appear in the paragraph but is implied in the context. The verb phrases that represent action in material process are “have prepared and organized”. It is worth noting that the person in charge utilizes “We” instead of “I” in the narration, which reflects that the narrator attributes the efforts made for the rescue work to the team to which he belongs. The “goal” of the action “have prepared” is the relief supplies, namely “bread, milk, bottled water and daily necessities”. The second verb “organized” refers to the goal “42 volunteers”, reflecting the important function of the person in charge to coordinate the deployment of materials and personnel in the context of disasters. This indicates that when flood disaster occurs, local departments unite as one and make outstanding contributions to ensuring the basic needs of the affected people. This not only allows readers to see outstanding individuals in disaster situations, but also to be moved by the kind and friendly team spirit.

In example 4, the actor of the material process is “aid agencies”. The verb phrase that expresses material process is “have been working to get relief supplies to”. The action is oriented towards the goal of “thousands of people in south-east Afghanistan, struck by an earthquake in late June”. The material process uses the present perfect progressive tense to show the relief agencies’ continued efforts to provide material support to the affected people. The phrase “in south-east Afghanistan” belongs to the “location” circumstance, which is placed after the goal of the action, that is, the “victims”, to supplement the area where the earthquake occurred. The circumstantial component is also represented by the past participle structure “struck by an earthquake in late June”, which implies the time of the disaster and the reason for the rescue. They serve to add details and enhance the credibility of discourse.

Example 5 contains two material processes. The first material process is initiated by the collective noun “rescuers”. The verb indicating the action is “had constructed”, and the goal of the action is “a temporary pedestrian bridge”. Their action provides convenience for the victims and facilitates rescue operations. The second action is initiated by “rescuers” again, the process is “used” and the goal is “ropes”. This is followed by three nested purpose adverbials, namely, “to rappel across the river, to reach stranded people, and move the injured to hospitals”. These processes illustrate the sense of responsibility of rescuers to actively assume the responsibility of restoring infrastructure in time and protecting the lives and health of the people after a natural disaster.

In example 6, the material process is initiated by the actor “the storm”. Moreover, the following four expressions representing “action” explain the severity of the storm in turn. The goal of the action “damaged” is “bridges and factories”; the goal of the action “torn off” is “roofs of buildings”; the goal of the action “triggered” is “widespread flooding and landslides”, and the goal of the action “leaving” is that “64 people still missing”. Through four material processes, this example shows that the storm severely damaged people’s homes and workplaces. These adverse results directed by the material process show a progressive meaning, highlighting the severity of the storm. This kind of repeated actions initiated by the same actor and a series of adverse consequences are more likely to stimulate the reader’s psychological reaction.

4.1.2 Verbal process

Verbal process refers to the situation in which a sayer expresses information through speaking. Verbal process involves the “sayer”, the “receiver” and the “verbiage”. The “sayer” is an institution or individual that expresses his views and feelings about the disaster. The “receiver” is usually the news readers or a reporter who interviews, so they will not be overemphasized in analysis. The “sayer” in natural disaster news can be divided into folk voices, official voices and media voices. Their discourse reflects their institutional identity, such as the formality and rigor of official statements and the emotional appeal of media discourse.

The verbal process is mainly used to describe casualties, escape experiences, rescue work and the public’s views on disasters. The verbal process is generally represented by “told, said, spoke to, declare, announce, explain” and their variants. The following examples are related to the overseas aid operations of the Chinese rescue team, the experiences of Chinese citizens during the Tonga volcanic eruption, and the Chinese government’s distribution of supplies during disasters. The analysis focuses on who is the speaker of the discourse, what event the discourse refers to, and to whom the discourse is addressed.

Examples:

7. This rescue work will demonstrate our professional rescue abilities and Chinese civilian rescue forces to promote international humanitarianism. *Luo Mingwen*, leader of the Chongqing Blue Sky Rescue Team, *said and added*, “I believe we can complete this mission.” (China Daily)

8. “The sky turned dark in minutes. Particles kept falling from the sky like hailstones, rattling against the windowpanes and blocking our sight,” *said Zhao*. “There was nothing we could do. We were panicking”. (China Daily)

9. Zhou Zhijun, head of the 15-strong team from the Red Cross’ Ningxia branch, said that four on his team were Red Cross personnel and the remaining were volunteers with full-time jobs. They also brought 500 tents, 500 folding beds and 4,500 jackets among other aid supplies.

(China Daily)

Analysis: Example 7 is about the news that China's Blue Sky Rescue Team participated in the local rescue during the Turkish earthquake. In this verbal process, the sayer is "Luo Mingwen", the leader of the Blue Sky Rescue Team. The word indicating verbal process is "said", which introduces verbiage "This rescue work will demonstrate our professional rescue abilities and Chinese civilian rescue forces to promote international humanitarianism." This shows the professional competence of the rescue team and China's kind humanitarian assistance to foreign countries. The second word indicating the verbal process is "added", which introduces verbiage "I believe we can complete this mission". It can be summarized as a mental process. Mental process describes the psychological state, such as worry, fear, satisfaction, anxiety, etc. It reflects people's subjective perception and attitude towards the event, and can be further divided into cognitive process and perception process. The identity of the person in charge changes from "sayer" to "sensor". The word "believe" is a marker word reflecting the cognitive process, and the perceived phenomenon is the working ability of the Chinese rescue team. It shows that the person in charge is full of confidence and determination in their rescue capabilities. Expressing positive emotions in the context of disasters can bring hope to the local victims and relieve their anxiety and panic. By conveying positive messages, official agencies can bring spiritual encouragement to people and enhance their confidence in dealing with disasters.

Example 8 is about "Mr. Zhao", a Chinese worker, describing his experience during the volcanic eruption in Tonga. The verbiage is "The sky turned dark in minutes" and "Particles kept falling from the sky like hailstones, rattling against the windowpanes and blocking our sight". The actor "particles" in the second clause perform actions such as "kept falling, rattling against, and blocking". These actions are directed to the goals such as "windowpanes" and "our sight", highlighting the terrible situation when the volcano erupts." Such words with striking and pointing characteristics highlight the terrible situation during the volcanic eruption. "There was nothing we could do" can be regarded as an "existential process" expressing negative meaning, indicating that the disaster event brought normal life to a standstill.

The description of the disaster witness allows readers to more deeply understand the terribleness of the disaster. The sentence "We were panicking" is an obvious mental process. The sensor "We" refers to the speaker and his colleagues, while the phenomenon, or the object being perceived, is "volcanic eruption". First-person narration enhances the audience's sense of involvement and implicitly classifies them as members of the group, thus establishing a channel of trust. This reflects the fear and worry of the people affected by the disaster, which is more likely to arouse emotional resonance in the listener.

As the representative of folk voices, the sayers are usually disaster victims, such as "survivors,

residents, witnesses, and local villagers”. By directly quoting the refugees’ personal accounts, the authenticity and destructiveness of the disaster are more intuitively presented to people, which has a stronger educational and warning effect. Folk voices are conducive to stimulating readers’ empathy and gaining readers’ attention and sympathy.

Example 9 is representative of the voice of non-governmental organizations. The sayer is “Zhou Zhijun”, head of the 15-strong team from the Red Cross. The verbiage consists of two parts, of which the first part is an introduction to the team members, which is a clause led by “that”. The second part explains the materials prepared by the organization through material process, such as 500 tents, 500 folding beds, and 4,500 jackets. Digital evidence provides precise, quantitative material conditions and is highly persuasive when combined with reliable sources. What they provide can alleviate local shortages and calm people’s fear, panic and anxiety. This has shaped the image of members of the social relief group “Red Cross” as being quick to respond, conscientious and hardworking in disaster situations.

Official voices account for a large proportion of disaster reports, and the main sayers are “government officials, emergency management agencies, disaster relief personnel, volunteers, etc”. Official statements are highly authoritative and credible, and their descriptions of the number of missing and injured people are more convincing. Verbal process also plays an emergency management role in guiding public opinion and clarifying rumors, such as a report from BBC News: “The military government has said that unofficial death tolls surpassing 400 are false”. The word “false” definitely confirms the falsehood of the rumor. This is a manifestation of the news media quoting government statements to correct erroneous statements.

Examples:

10. I heard the thunder and the sound similar to boiling water. It was the sound of big rocks falling from Mount Marapi,” *Rina Devina*, a 43-year-old housewife from the Agam district *told AFP news agency*. (BBC News)

11. One villager, *Taj Ali Khan*, *said* there are also serious shortages of medicines, equipment and health workers to treat those injured. (BBC News)

12. *I told them* “we have to get out now as I watched the ground crumble towards the back door. I was petrified.” (BBC News)

Analysis: In the verbal process of example 10, the sayer is “Rina Devina, a 43-year-old housewife from the Agam district”, who has witnessed the disaster. The verb that indicates the verbal process is “told”. The recipient is “AFP news agency”. The verbiage consists of a mental process and a relational process. As for mental process “I heard the thunder and the sound similar to boiling water, the sensor is “I”, and the phenomenon is “the thunder and the sound similar to boiling water”. “Heard” is a word that indicates the mental process, indicating that “I felt a certain

phenomenon through hearing”, namely “thunder and a sound similar to boiling water”, which are some signs before the flood.

Here the rhetorical effect in the relational process is briefly explained. The “carrier” usually represents an entity with certain characteristics, while the “attribute” is the description and explanation of the characteristics of the carrier in relational process. In this relational process, the carrier is “it”, which refers to the thunder mentioned earlier, and then the attribute is “the sound of big rocks falling from Mount Marapi”. The speaker uses the rhetorical device of simile here, comparing the sound of the big rocks rolling down from the top of the mountain to boiling water, which arouses the reader’s auditory association. This helps readers use the familiar concept of “boiling water” to understand the rumbling sound of the rocks rolling down, reflecting that news discourse has certain literary and aesthetic characteristics.

In the verbal process of example 11, the sayer is “One villager, Taj Ali Khan”, and verbiage is “there are also serious shortages of medicines, equipment and health workers to treat those injured”. This verbiage can be summarized as an existential process, in which the “existent” is “shortages of medicines, equipment and health workers to treat those injured”. This process speaks out the problem of shortage of medical supplies and medical staff in the context of disasters from the perspective of the villager, who is also a survivor of the incident. This is conducive to attracting the attention and donations of all sectors of society by leveraging the rapid dissemination power of the media.

In the verbal process of example 12, the sayer is “I”, the word indicating the verbal process is “told”, the recipient is “them”, and verbiage is “we have to get out now as I watched the ground crumble towards the back door”. and “I was petrified.”. This first-person narrative has a more realistic style, allowing readers to stand in the perspective of someone who has experienced the disaster and feel his tense emotions at the time. Verbiage contains two processes. In the first mental process, the perceiver is “I”, the word representing the mental process is “watched”, and the perceived phenomenon is “the ground”. The circumstantial factors that indicate the manner and direction are “crumble” and “toward the back door” respectively. It shows the horror scene of the flood. In the second mental process, the sensor is “I”, and the word that represents the mental process is “was petrified”, which is the speaker’s psychological reaction to the collapse of the house. This mental process reflects the extreme fear of the victims in the face of the storm.

4.1.3 Relational process

According to the definition of relational process in transitivity, relational process mainly describes the state or relationship between things. There are characteristics and identity relationships, relative position relationships, and relationships of existence or non-existence

between things. Relational process includes two categories “attribution” and “identification”, that is, generalizing things into a certain type or identifying the features of things (Hu et al., 2005). In the “identifying” mode, some thing has an identity assigned to it, which means that one entity is being used to identify another (Halliday, 2004:227). The two participants in the attributive relational process are the “carrier” and the “attribute”. However, in the identifying relational process, the participants are the “identified” and “the identifier”. The “identified” is the person or item with characteristics recognized by the subject, while the “identifier” is related to attributes and features.

In natural disaster news, relational process is mainly used to present material supply, the cause of the event, the characteristics of the disaster, and science knowledge of emergency. Typical words that may reflect relational process are “is, has, serve as” and their various tense variants. The following examples are related to weather warning systems, river dike construction, Southern California fires, and typhoon knowledge popularization. The examples will be divided into attributive relational process and identifying relational process, and their characteristics and structures will be clarified.

Examples:

13. *China has a four-tier weather warning system, with red representing the most severe warning, followed by orange, yellow and blue.* (China Daily)

14. *The dike breach should serve as a lesson for all to prepare for emergency response in the future.* (China Daily)

15. *Bridge Fire is one of three major fires in southern California, along with Line Fire and Airport Fire.* (China Daily)

16. *“Our house was up there,” he says pointing to the area where it once stood. It is now just part of an expanse of debris.* (BBC News)

17. *A typhoon is a deep area of low pressure that forms from an organized cluster of thunderstorms over warm tropical waters.* (BBC News)

18. *Human-caused warming was also the main driver of the Amazon rainforest’s worst drought in at least half a century in the second half of 2023.* (BBC News)

Analysis: Example 13 is a process that expresses the relationship of “attribution”. The carrier is “China”, which is an entity with specific properties. The attribute is “the four-tier weather warning system”. In the independent participle clause introduced by “with”, the carrier is the “red in warning system” and the attribute is “the most severe warning”. In this example, the words indicating the relational process are “has” and “representing” respectively. It explains China’s four-tier weather warning system, especially emphasizing that “red” is the most severe warning signal, which is a reminder of the dangerous weather conditions. It is a manifestation of the role of

language resources in popularizing emergency knowledge. Example 14 belongs to an attributive relational process. The carrier is “dike breach”, an entity with certain features, the relational process is expressed by “should serve as”, and the abstract attribute is “a lesson”. This shows that the news has a warning effect on people, prompting relevant departments to strengthen the dike to prevent natural disasters. Example 15 is a typical identifying relation, in which the process word “is” connects the identified “bridge fire”, and the identifier “one of the three major fires”. The circumstantial factor is played by the adverbial of place “in southern California”, determining the geographical location.

Example 16 contains two identifying relational processes, reflecting the sharp contrast between people’s residence before and after the disaster. In the first sentence, “our house” is the identified, the word indicating the relationship is the past tense verb “was”, and the identifier is “up there”. In the second sentence, the identified is “it”, which refers to our house mentioned earlier. The word indicating the relationship is the present tense “is”, and there is a very clear environmental factor “now”. The identifier “just part of an expanse of debris” describes the state of the house after the disaster. Through this comparison before and after the disaster, the news very intuitively presents the serious damage caused by the disaster to people’s homes.

Examples 17 and 18 are both identifying relational process, because they both have a phenomenon that still holds true after inversion. Example 17 is an example used for popularizing disaster knowledge, describing the principle of typhoon formation. The identified is “A typhoon” and the word indicating the relational process is the word “is”. The identifier is “a deep area of low pressure that forms from an organized cluster of thunderstorms over warm tropical waters”, indicating the climate-related conditions for typhoon formation. Then we reverse the identified and the identifier, we can get “a deep area of low pressure that forms from an organized cluster of thunderstorms over warm tropical waters is a typhoon”. This has the same meaning as the original sentence, so it belongs to the identification category. In example 18, the identified is “human-caused warming”, the word indicating the relational process is “was”, and the identifier is “the main driver of the Amazon rainforest’s worst drought in at least half a century in the second half of 2023”. If we reverse the order of the identified and the identifier, the meaning of the sentence is the same as the original text, so it is consistent with the characteristics of the identifying relationship process. It describes that human factors are important causes of climate warming, reminding people to reduce improper human behavior that damages the environment to curb climate warming.

4.2 Interpersonal Function of News Discourse

Interpersonal function is expressed through the modality system and the mood system. The

analysis of modality is based on Halliday’s three-level classification of modal word values. Modal words of different levels will affect the certainty of the event referred to by the discourse and imply the speaker’s position. The mood system refers to the speaker’s tone, such as using exclamatory tone to express surprise, fear, etc. However, the expression of tone in news texts is not obvious, and it is mainly manifested through punctuation marks. Below the writer explains the classification of the modality system and its manifestations, and the severity of events reflected by the tone.

4.2.1 Modality system

The modality system reflects the sayer’s judgment on the possibility of the topic, as well as his responsibility and attitude towards the content of the discourse. Table 4.3 shows the classification of modal words. High-value modal words reflect strong certainty and express a firm position. Medium-value modal words express the user’s relatively neutral attitude and euphemistically express the uncertainty of the topic. Low-value modal words reflect strong uncertainty, leaving a certain amount of room for discussion on the topic, and are most likely to gain the listener’s approval. Different tones, such as declarative, interrogative or imperative, can realize different discourse roles and reflect the social relationship between communicators (Ma et al., 2017). The writer utilizes corpus retrieval software Antconc 3.5.9 to search for modal words in natural disaster news, and the results are shown in Table 4.3

Table 4.3 Frequency and proportion of various modal words

Modal Values	Modal Words	China Daily		BBC News	
		Frequency (times)	Proportion %	Frequency (times)	Proportion %
High-value	must, ought to, need, had to, have to, has to	53	24.31	50	15.33
Medium-value	will, would, should, is to, was to, will not, would not	71	32.56	122	37.42
Low-value	can, may, could, might	94	43.11	154	47.23

Overall, among the collected corpora, the modal words used by BBC News are 108 more than those used by China Daily. As can be seen from the Table 4.3, low-value modal words account for the highest proportion in China Daily and BBC News, 43.11% and 47.23% respectively. Among them, “can” and “could” are used most frequently. Medium-value modal words account for the second highest proportion, 32.56 % and 37.42% respectively. Among them, “will” and “should” account for the highest proportion. The main reason is that the development of natural disasters is changeable and may trigger other disasters, and it may be difficult to collect clear and detailed

information in the early stage due to damage to infrastructure and far distance. Therefore, the media tends to maintain a cautious and objective attitude and offer a reserved narrative.

High-value modal words account for the lowest proportion. “Need” and “have to” are used more frequently in relevant reports. The reason is that these modal words are too imperative and deterministic, which is inconsistent with the objectivity of news discourse. However, the proportion of high-value modal words in China Daily is 8.98% higher than that in BBC News. The function of modal words in natural disaster news is to emphasize the determination and ability of rescue, guess the causes and trends of disasters, calling for rescue supplies and manpower support. Of course, they are also used to reduce the speaker’s speculation about the possibility of the situation, express uncertainty, and avoid a sense of obligation and responsibility. The following is an analysis of the interpersonal function of news discourse.

Examples:

19. China’s top policymakers called on Thursday for unrelenting efforts in flood control and prevention, stressing that the safety of people’s lives *must always be* the top priority, and all-out efforts *must be* made in rescue and disaster relief. (China Daily)

20. Such a situation *could easily trigger* waterlogging or flooding in provinces that were previously experiencing drought, such as Hubei, Anhui and Jiangsu. (China Daily)

21. During our retreat to our base at a higher altitude shortly after the eruption, we *could not see* anything even with our high beam headlights on. (China Daily)

22. Even if tap water is filtered, the CDC guidance states that it will still *need to be* boiled as it *may be* contaminated. (BBC News)

23. Polish Prime Minister Donald Tusk said one billion zloty *would be* allocated for flood victims in the country, adding that Poland *would also apply for* EU relief funds. (BBC News)

24. Rising temperatures *may also* increase the likelihood of lightning in the world’s northernmost forests, triggering fires. (BBC News)

Analysis: Example 19 uses two high-value modals, namely “must always be” and “must be”. High-value modal words indicate strong definite attitude and have strong emotional appeal. These two words are used with “the top priority and all-out efforts” respectively, and are accompanied by the intensifier “always” to enhance the absoluteness of the proposition. This shows that in the context of flood prevention and control, the Chinese government emphasizes the concept of putting people’s lives first and has a sense of responsibility to do its utmost in the rescue process. In example 20, a low-value modal word “could easily” is used to express uncertain speculation, which is adopted here to infer the cause of heavy rain. It indicates that officials are relatively uncertain about the possibility that heavy rains cause flooding in areas that have experienced drought. This is unlikely to attract the listener’s attention or alertness, yet it provides residents with early warning

information that flooding may be imminent.

In example 21, the low-value modal “could not see” with negative meaning does not have an inferential effect here. It describes that the witnesses lost the ability to see things clearly during the volcanic eruption. The narration by eyewitnesses of the disaster makes readers feel as if they are experiencing the dark scene of the volcanic eruption, causing them to care about and sympathize with the narrator. In example 22, the main clause of the concession adverbial clause uses a high-value modal verb “need” and a low-value modal verb “may be”. “May be” indicates that the news release agency, “the Centers for Disease Control and Prevention (CDC)”, is uncertain about whether the water source is contaminated. However, out of consideration for protecting people’s health, the agency used the high-value modal word “need” to indicate a stronger certainty, emphasizing that the boiled water is more hygienic.

Example 23 contains two medium-value modal words, namely, “would be allocated” and “would apply for”. They indicate the speaker’s relatively neutral attitude. The subject of “would be allocated” is “one billion zloty”, and the object of “would apply for” is “EU relief funds”. The beneficiaries of these two actions are the “flood victims”. These two modal verbs express an objective statement that the victims of the flood will receive financial assistance, which is a matter of vital interest to the people. In fact, if high-value modal words are used, it would be easier to inspire people’s confidence in the disaster context. Medium-valued modal words are less likely to arouse people’s belief, but are more likely to make people doubt or speculate. Example 24 uses the low-value modal verb “may increase”, which means that “lightning caused by high temperatures” may be the cause of the forest fires in the northern region. This is a speculation about the uncertainty of the causes of forest fires. This implicit and euphemistic statement illustrates the speculator’s rigorous and objective attitude, thereby arousing people’s vigilance and awareness of prevention.

4.2.2 Mood system

The noun “mood” is generally believed to be derived from the Latin and French mode, which originally means the way of speaking (Shi, 2024). The mood system can be expressed through interrogative sentences, imperative sentences and exclamatory sentences. It enables the audience to sense the speaker’s emotions, such as nervousness, calmness, and delight, and to grasp the speaker’s intentions, such as praise and request. The difference in mood in text is vaguer, while videos in disaster news can better reflect the speaker’s emotions and attitudes through mood. However, there are not many cases related to mood in both corpora, so they are not counted. The writer has selected some typical examples to illustrate how to achieve interpersonal interaction through the mood, that is tone.

Examples:

25. “The flood is coming, get up quickly!” shouted Xiang Quanxin, a local volunteer, while knocking on the doors from house to house. He helped inform over 20 households to evacuate. (China Daily)

26. “I am very worried ... My God! It’s terrible.” resident Andrea Gomez, 48, said. (China Daily)

27. “How do you rebuild it? This is complete devastation,” said Kelly Trussell, a lifelong Sulphur resident as she surveyed the damage. “It is crazy, you want to help but where do you start?” (China Daily)

28. If you are seeking warmth in your car, remember the fumes coming out are toxic, so always have a window open and never switch on your engine in a closed garage. (BBC News)

29. “I’m in Houston, Texas freezing to death,” one Twitter user, Chris Prince, wrote. “No power, no heat, no water. I have four young children. How is this happening right now?” (BBC News)

30. “If you were here, you would cry instantly, because people are desperate, their whole lives’ work is gone, there were people who were left with just the clothes they had on,” he said. (BBC News)

Analysis: Example 25 adopts a series of material processes to illustrate the urgent situation at the disaster site. The verbs that express the material process are “is coming, get up, shouted, knocking, helped inform”. The speaker is a person who experienced the flood disaster and is also a volunteer who participated in the evacuation. Here, an exclamatory tone is used “the flood is coming, get up quickly!”, establishing a tense atmosphere and are more likely to attract the audience’s attention and emotional response. As an imperative sentence, “get up quickly” is more concise and powerful, highlighting the urgency of the situation. This reflects the community workers’ urgent mood at the time to protect the lives and health of the people. In example 26, a forest fire broke out in the area where the speaker, resident Andrea Gomez, lived. The fire is so strong that it endangers the place where she lives. She said to the camera, “I am very worried ... My God!”, revealing her mental process as “worried”. The exclamatory tone shows her shock and fear. A foreign idiom like “My God” can resonate with the listener more, eliciting their care and compassion.

There are two interrogative sentences in example 27, which make the listener subconsciously assume that he or she is the one who needs to give an answer, thus attracting readers to participate in discourse construction. Resident Kelly Trussell asks the audience “how do you rebuild it”? The pronoun “It” refers to the fact that tornado has destroyed many surrounding buildings. Faced with the devastated post-disaster scene, Kelly Trussell shows a lack of confidence and doubt about the

reconstruction of her living area. In the second interrogative sentence, “You want to help, but where do you start?”, the first half assumes that the listener is willing to lend a helping hand, and the second half, “but where do you start?”, indicates a semantic shift. The speaker questions the possibility of helping the local people rebuild, indicating that she seems to have lost hope in restoring the local normal living environment. Such interrogative sentences are likely to trigger the audience’s thinking and arouse their sympathy.

Example 28 is a conditional adverbial clause that first anticipates a possible situation and then gives a clear solution. The condition is “If you are seeking warmth in your car”, which indicates the possible actions that the listener may take under certain conditions. The following three imperative clauses provide safety measures that should be taken, namely “remember the fumes coming out are toxic, have a window open, and never switch on your engine in a closed garage”. Imperative sentences with omitted subjects allow the listener to find key information more quickly. It serves as a disaster warning to people, providing important information about the toxicity of car exhaust and the safe use of car heating. By giving advice or warnings, a closer psychological relationship is established with the reader, expanding the audience’s emergency knowledge reserves.

In example 29, the speaker, Chris Prince, first uses the indicative mood to describe his unfortunate experience during the winter storm. Then she utilizes the question “How is this happening right now?” to express her surprise and helplessness about the storm. Interrogative sentences can be seen as the speaker’s attempt to communicate with the listener, which can easily attract the listeners’ attention and sympathy. Research shows that there are a large number of interrogative sentences in BBC news, such as “How powerful was the storm? Is climate change to blame? Is aid reaching earthquake victims? and interrogative sentences are often used as news titles. News writers usually put these questions in bold to make them more eye-catching. The headlines of news in China Daily are usually declarative sentences. This is also a significant difference between the two media, which is influenced by Chinese and Western values and media communication styles. Chinese news serves the traditional concept of harmony and stability, so it advocates a conservative and objective understatement, while Western media focuses on promoting freedom and critical thinking, so the news language is more intense and contagious.

Interrogative sentences express inquiries that have not yet reached a conclusion and are more likely to trigger the audience’s thinking to the issue. It even has some effect of condemning and expressing doubt. In the sentence “Why are we planning to build more homes in high-risk flood areas when we know that the situation is only going to get worse because of the climate emergency? (BBC News)”, the interrogative sentence indicates the speaker’s attitude of condemnation and dissatisfaction. It condemns the behavior of municipal planning departments to build housing in

high-risk areas, prompting relevant parties to change inappropriate measures.

Example 30 uses the subjunctive mood, which can enhance the appeal of the speech. The subordinate clause “if you were here” seems to bring the reader to the real disaster site, indicating a situation that does not conform to the current situation. The main clause is “you would cry instantly”, imagining the situation that might happen to the listener. The empathy effect refers to the ability to understand other people’s situations through watching or imagining, and to have similar emotional responses (Fu et al., 2017). By constructing a fictitious scenario to express the speaker’s emotional attitude, this can stimulate the reader’s empathy effect. It allows readers to imagine or feel the sad atmosphere brought about by the disaster. At the same time, it uses direct quotations to make the discourse more real and vivid, thus enhancing the credibility of the information.

4.3 Textual Function of News Discourse

Textual function refers to how various language elements in a text are combined to convey meaningful information. Halliday (2004) believes that textual function is manifested by theme-rheme structure, information structure and cohesion. The following is an explanation of the criteria for determining marked and unmarked themes and strategies of discourse cohesion. This paper will analyze how news can highlight the main information and make the logic clear by changing the position of information and adopting different strategies.

4.3.1 Theme-rheme structure

The Theme is the element which serves as the point of departure of the message, which can be divided into marked theme and unmarked theme (Halliday, 2004:64). We shall refer to the mapping of theme on to subject as the unmarked theme of a declarative clause (Halliday, 2004:73). Marked theme means that the theme placed at the beginning of the sentence is different from the subject. The rheme is the detailed information surrounding the theme, which is usually the key content of the discourse. According to Halliday, the theme is usually known information, while the rheme is the new information that the sentence wants to provide. Halliday (2004:79) divides the theme into topical theme, textual theme, and interpersonal theme. The textual theme connects the sentence with the rest of the text. (Zuo, 2007). The topical theme refers to the theme component that is closely related to the theme of the sentence. The interpersonal theme has the function of interacting with the third party and expressing the speaker’s emotions and attitudes. The researcher annotates and proofreads the corpus respectively with the help of the theme system in Uam Corpus Tool and obtains the following results:

Table 4.4 Frequency and proportion of thematic categories in natural disaster news

Thematic Category	China Daily		BBC News	
	Frequency (times)	Proportion (%)	Frequency (times)	Proportion (%)
Unmarked Theme	1079	66.48	1346	69.49
Marked Theme	544	33.52	591	30.51
Topical Theme	1548	85.54	1863	83.73
Textual Theme	267	14.58	324	14.56
Interpersonal Theme	16	0.87	38	1.71

As can be seen from Table 4.4, the proportion of unmarked themes in China Daily and BBC News is 66.48% and 69.49% respectively. This is much higher than the marked themes (33.52% and 30.51% respectively). The proportion of topical themes in China Daily is slightly higher than that in BBC News. The proportion of textual theme in China Daily and BBC News is roughly equal, 14.58% and 14.56% respectively. However, the proportion of interpersonal themes in China Daily is lower than that in BBC News. News articles are usually short and timely, and unmarked themes and topical themes are more in line with normal word order and the reading habits of the general public. They allow readers to quickly obtain information in a concise paragraph, while other types of themes have too many modifiers and flexible positions, requiring readers to respond for a longer time. The following is a comparative analysis of the theme-rheme structure.

Examples:

31. Fortunately, all the villagers are safe,” he said. (China Daily)

32. A team of 20 firefighters from the Liangshan prefecture team were carrying out the search operation with a search and rescue dog and portable breaking tools. (China Daily)

33. On Sunday, the mayor of Klodzko announced that his town had “lost the battle” against the floods, and that the situation had become “critical”. (BBC News)

34. Unfortunately, no one knows when the water will recede. (BBC News)

Analysis: Example 31 is a typical marked theme, because the theme “fortunately” is placed at the beginning of the sentence to play an emphasis role, while the subject of the sentence is “all the villagers”. The theme of the sentence is not equivalent to the subject, which meets the definition of a marked theme. It is also an interpersonal theme, because “fortunately” conveys a positive signal and expresses the narrator’s attitude towards the event. It makes a positive assumption for the development of the event, allowing readers to associate it with a better result when they see the beginning. The rheme “are safe” indicates that the villagers are safe and healthy, which just confirms the reader’s speculation.

Example 32 belongs to an unmarked theme. Because the subject of the sentence “A team of 20 firefighters from the Liangshan prefecture team”, is placed at the beginning of the sentence and

is also the theme of the sentence. That is to say, the subject and the theme are played by the same content, so it belongs to an unmarked theme. Furthermore, it belongs to the topical theme. The reason is that “a team of 20 firefighters from the Liangshan prefecture team” is the center of the sentence, that is, the rest of the components revolve around supplementing and explaining it. Moreover, the entire rheme “were carrying out the search operation with a search and rescue dog and portable breaking tools” is developed around it. The rheme explains in detail the searching and breaking tools used by the rescue team in their search effort.

Example 33 belongs to the marked theme, in which the theme is “on Sunday”. The sentence does not put the time adverbial “on Sunday” at the end of the sentence where it is usually placed, but at the beginning of the sentence, that is, the theme position, indicating that the speaker wants to highlight the specific time when the event occurred. At the same time, it is a time component in topical theme because the rheme of the sentence “announced that his town had failed in the battle against the flood and the situation had become critical” is a comprehensive presentation of the mayor’s speech on Sunday. It contains the metaphor of “responding to floods is a battle”, in which the source domain is “flood” and the target domain is “war”. The “participants in flood control” can be mapped to “soldiers” in the target domain “war” because they are living participants in the corresponding activities. “The prevention and control measures of floods” form a mapping relationship with the “combat strategy” of the “war” because they belong to skills in different fields. “The characteristics of floods with strong destructive power” can be mapped to “the features of war causing loss of life and property”. This cross-domain implicit relationship shows that floods are rampant and destroy people’s lives, reflecting people’s strong and brave character in dealing with disasters.

Example 34 belongs to a marked theme and an interpersonal theme because the theme of the sentence “Unfortunately”, is placed at the beginning of the sentence, not the subject “no one”. The negative interpersonal theme “Unfortunately” shows that the narrator is skeptical and disappointed about the event. It presupposes a bad ending in the readers’ mind for the event to be narrated, causing them to feel depressed. The readers can associate it with a bad outcome when they see the theme at first. The predicate “know when the water will recede” combined with “no one” here proves that people do not know when the flood will recede. The meaning of the rheme also confirms the sad and negative tone conveyed by the theme.

4.3.2 Information connection and coherence

Coherent news discourse can reduce the difficulty of understanding, enabling readers to master key information easily and smoothly, and demonstrating the superb professional quality of practitioners. Cohesion refers to the connection between the meanings of various language units,

including lexical cohesion and grammatical cohesion (Halliday, 2004). The connectives in the two corpora can be roughly divided into causal connectives, temporal connectives, coordinating connectives, etc. They help to smoothly introduce topics, achieve semantic conversion, and make the logic of news clear. The following selected examples further illustrate the theme-theme structure and discourse coherence.

Examples:

35. Additionally, over 1.1 million people were urgently relocated, with direct economic losses amounting to 76.8 billion yuan. (China Daily)

36. Thanks to the rescuers, the residents in Wuqiao were evacuated in an orderly manner and sheltered in safety. However, Zhang Shenglan, a 78-year-old woman who lived alone, was trapped inside her house as the floodwaters rose. (China Daily)

37. The world has already warmed by about 1.1C since the industrial era began and temperatures will keep rising unless rapid efforts to curb greenhouse gas emissions are made. (BBC News)

38. As a result, power supply has been cut off, and because of that telecommunication systems were also not functioning.” (BBC News)

Analysis: The adverb “Additionally” in example 35 is a coordinating connective and a marked theme. The reason is that “additionally” is placed at the beginning of the sentence and it is different from the subject “over 1.1 million people”. It can be seen that the subject and the theme are played by different components, so it belongs to a marked theme. The rheme is “urgently relocate”, which is a detailed explanation of the theme, indicating that the affected people have been resettled in a timely and proper manner. The word “Additionally” introduces a supplement to the previous content, namely “the number of people who died or went missing due to disasters in July”. This makes the information richer, and the subsequent information on displacement and economic losses shows the proactive actions of the management sectors and the severe impact of the disaster.

In example 36, “Thanks to” is a logical conjunction that indicates a causal relationship. It usually refers to positive factors that lead to favorable results. Here, it is a marked theme with positive semantics, indicating that the local residents have been safely evacuated. The word “However”, indicates a semantic transition, suggesting that the event is developing in a negative direction. It leads to the fact that an elderly woman is trapped. The combination of these two logical conjunctions makes the discourse more compact and coherent in meaning. It is common to first describe positive and inspiring results in news texts, and then add negative and sad events. This gives readers a certain amount of thinking buffer time, so that the whole experience of obtaining information will not make people feel too frustrated.

In example 37, there are two conjunctions “since and unless”. The word “since” introduces a time adverbial clause, indicating that something continues from a specific time point in the past to the present. Here, it refers to the fact that the temperature has risen by one degree since the industrial revolution. The word “Unless” in the second half of the sentence, indicates that the result can only be achieved if certain conditions are met. Here, it means that only by curbing greenhouse gas emissions can the rise in temperature be slowed down. These two conjunctions connect the industrial revolution, rising temperatures, and greenhouse gas emissions together, allowing readers to better understand the time and causal relationship between them.

In example 38, “as a result” is a causal conjunction and a textual theme, referring to the result of the development of the event. Considering that people tend to focus on the impact of disaster events, the media uses “as a result” as the theme to guide the infrastructure failure caused by the flood. Therefore, it is also a textual theme, making the discourse more coherent and organized. Common textual themes include “however, meanwhile, and at the same time”. The textual theme can reflect the intertextuality of the text and enhance the relevance of the content. The prepositional phrase “because of” supplements the adverse effects of the failure of the telecommunications system due to the cut-off of electricity, enabling readers to have a more comprehensive understanding of the series of adverse consequences brought about by the disaster.

Through the above examples, the vocabulary, sentences and perspectives used by Chinese and English news in describing disaster events are presented. The differences shown by the statistical results and their possible reasons are also discussed. Overall, from an objective third-person perspective, the news shows that social institutions and people are united in rescue and reconstruction under the background of disasters.

5. Comparative Analysis of Natural Disaster News Images

News images are important carriers for presenting facts and promoting emotional connections. Diverse and interesting image symbols make news more vivid and even reduce the comprehension obstacles caused by cultural differences and cognitive limitations. Based on visual grammar, the writer annotates and counts the elements of the representational meaning, interactive meaning, and compositional meaning of image resources in natural disaster reports. This research will conduct case analysis on the emergency management effects of picture elements such as demanding image, distance, contact, and perspective, etc. The following is the distribution of each element and its specific explanation.

5.1 Representational Meaning of Images

Images give people a more intuitive visual experience than words, and the combination and arrangement of elements has an important influence on the dissemination of meaning. Vector divides the representational meaning into narrative representation and conceptual representation (Kress & Van Leeuwen,1996: 62). Narrative representation includes action process, reaction process, speech and mental process (ibid: 63). Conceptual representation includes classification process, analysis process and symbolic process (ibid: 96). The images of these processes have different forms of expression and typical signs. According to the theory of visual grammar, the proportion of two representation forms in China Daily and BBC News are obtained after annotation.

Table 5.1 Frequency and proportion of conceptual representation and narrative representation				
Representational Meaning	China Daily		BBC News	
	Frequency (times)	Proportion (%)	Frequency (times)	Proportion (%)
Conceptual representation	41	51.25	27	32.53
Narrative representation	39	48.75	56	67.47

As shown in Table 5.1, the proportion of conceptual meaning of images in China Daily is higher than that in BBC News. Moreover, the proportion of narrative representation of images in BBC News is higher than that in China Daily. It can be seen that China Daily pays more attention to showing the static natural disaster scene, while BBC News pays more attention to reflecting dynamic events such as on-site rescue and material distribution. Typical pictures are selected and

analyzed below in terms of the specific elements of narrative representation and conceptual representation.

5.1.1 Conceptual representation

The participants in conceptual pattern have a stable and timeless nature in their structure or meaning (Kress & Van Leeuwen,1996:59).We have already discussed in Chapter 3 the manifestation of vectors, which is a real or virtual symbol that can direct one's attention to a particular directional space. There are no vectors in the conceptual representation, which usually manifests as a stable picture.



Figure 5.1 An aerial drone photo taken on July 28, 2024, shows rescuers relocating residents at Jingtang village of Zixing city,



Figure 5.2 Aerial view of the lava flow during the Mount Nyiragongo volcanic eruption near Goma, in the Democratic Republic of Congo May 23, 2021.



Figure 5.3 The world is on the brink of “catastrophic warming”. (China Daily, Zheng Huawei, April 10,

Based on the presence or absence of vectors in the image, Figure 5.1 can be judged as a conceptual representation, because there are no significant vectors such as the characters' body movements or eye contact in it. It just objectively presents the situation of houses being submerged in floods. This type of photo lacks dynamism and is not likely to evoke the emotions or attention of the audience. The classification process connects the participants with each other in a “kind of” relation, in which the members placed together are judged to belong to the same class (Kress & Van Leeuwen,1996:79). This picture belongs to the classification process, including five types of elements: houses, floods, field ridges and trees.

Figure 5.2 reproduces the eruption of a volcano, where the lava flow produced a large amount of black smoke in the air, occupying three-quarters of the picture. The basic elements in this picture are houses, engineering facilities, and smoke, all of which are static. It can also be identified as a conceptual representation because there are no vector-related signs in the picture. The image is intended to illustrate that the lava smoke gathers from all directions above the city, preventing the audience from seeing the disaster scene below. When the audience sees this picture, they realize

the intensity of the volcanic eruption and are eager to know what kind of suffering the people in the city shrouded in smoke have experienced.

Figure 5.3 is a cartoon from China Daily, which can be judged as conceptual representation and further subdivided into symbolic process. It refers to the use of abstract symbols to refer to events and convey information in an implicit way, requiring readers to use existing knowledge to imagine. The sphere in the center of the picture and the cylinder in the lower right corner are combined to form the shape of ice cream. The blue and green sphere symbolizes the earth, while the five points on its surface that droop downward symbolize the melting of ice cream. The grid on the brown column is very similar to building bricks, conveying the warning significance that climate warming will cause glaciers to melt and may flood human living space. At the bottom and right of the picture are twisted and entangled black curves, which converge respectively at the lower left and upper right corners to form four fingers, creating an uncomfortable sense of control. “Black” often symbolizes bad things, and here it can be associated with the adverse effects of various environmental damage activities that spread invisibly, eventually leading to a serious ecological crisis on the earth.

5.1.2 Narrative representation

Narrative representation presents unfolding actions and events, changing processes, and transitory spatial arrangements (Kress & Van Leeuwen, 1996:59). There is an obvious vector in Figure 5.4, which belongs to narrative representation and can be further summarized as the action process. Kress & Van Leeuwen (1996:63) hold that the actor is the originator of the vector, or it constitutes the vector itself (in whole or in part) in action process. Transactional process can be summarized as interactions between an actor and a goal, or between multiple actors (ibid: 64). In this picture, the actor is the firefighter, the sender of the vector, and the vector is his body movement of waving the plant violently. The final target of the “vector” is the rising white smoke, which is the recipient of the action.

According to the definition mentioned above, it can be seen as a transactional process, because there are two participants in the picture and the actor in white clothes swings the branch towards the goal “flame under smoke”. The man in the picture has a determined gaze, slightly squats down, and holds the plant stem firmly with both hands, as if he is ready to extinguish the smoke. Combining his expressions and large-scale movements, viewers can sense his eagerness and determination to put out the fire. This kind of image has a stronger sense of the power in action and the directionality of the state of motion.



Figure 5.4 People try to extinguish a forest fire in Nemocon, Colombia, on Thursday.

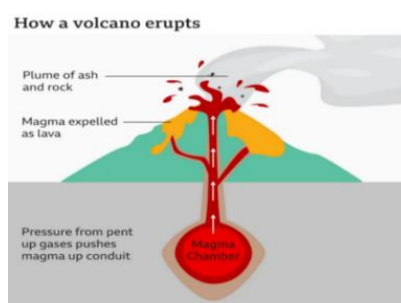


Figure 5.5 Magma comes out from “rifts”.



Figure 5.6 Xi urges “all-out effort” to protect lives in flood prevention.

Example 5.5 is a picture that aims to popularize emergency knowledge, and there are four arrows in it. BBC News released this picture with the purpose of demonstrating the process of volcanic eruption to the public, which is more in line with the goal of popularizing emergency knowledge. It can be judged as a typical narrative representation picture, in which the initiator of the action is “magma”. There are very obvious vectors in the picture, namely “four upward arrows” that guide people’s eyes sights from bottom to top. The goal of the action is “magma expelled as lava”. Combined with the text in the picture, readers can clearly understand the formation process of volcanic eruption. This kind of picture that can trigger dynamic visual changes in viewers reflects the role of image symbols in popularizing knowledge in natural disaster news. Such pictures used for safety science popularization require practitioners to spend more time and energy on the production of works, so that the complex causes and transmission routes of disasters can be presented to readers in a concise way. This demonstrates the sense of social responsibility and excellent working ability of media practitioners.

Figure 5.6 is a picture of firefighters rescuing people during the flood in Henan Province. In this picture, the rescuer in orange clothes on the left and the rescued people in green clothes on the right stretch out their hands and hold each other tightly. Their outstretched arms and clenched right hands form a distinct horizontal line, namely, vector. This bridge-shaped vector symbolizes the Chinese people’s determination to unite and bravely fight the disaster together. Combined with the slogan “Rescue on the way” above the picture, we can feel the urgency of the disaster and people’s joint efforts in that moment. This is a typical example of news organizations using image metaphors to showcase profound connotations such as people’s unity and friendship, making the pictures visually striking and emotionally appealing.

It is found that the way Chinese and English media present information through pictures is relatively monotonous and lacks innovative and attractive forms. Kress & Van Leeuwen (ibid: 68) argue that verbal and mental processes connect the speech or thoughts of the participants in the

image and readers through “thought bubbles or dialogue bubbles”. The arrow of the “thought bubble or speech bubble” is the vector of the process. However, none of the images related to verbal and mental processes appeared in the collected corpora. Therefore, it is suggested that news can introduce this form of information presentation, integrating elements such as cartoons or animations to reduce people’s visual fatigue. For example, practitioners can draw cartoon character propagandists to reveal first aid measures through interactive dialogue bubbles to stimulate the audience’s sense of participation.

5.2 Interactive Meaning of Images

The interactive meaning of the image focuses on the visual and psychological impact it brings to people, allowing the audience to perceive the emotions of joy, anger, sorrow and fear conveyed by the picture. It corresponds to the interpersonal function of language. To interpret the interactive meaning, it is necessary to comprehensively consider the relationship between the characters in the image, the news readers and the news creators. Wei (2012) points out that “lens language” refers to a language method in which the photographer uses a combination of film, television images and symbols to express things. The psychological reaction that the picture brings to the viewer, the viewer’s sense of participation in the scene of the picture, and the perception of the details are all interactions between the viewer and the image. The researcher establishes a labeling system based on visual grammar and selects the corresponding positions in the picture, and obtains the data as shown in Table 5.2.

Table 5.2 Frequency and proportion of elements in interactive meaning of images					
Interactive Meaning	Categories	China Daily		BBC News	
		Frequency (times)	Proportion (%)	Frequency (times)	Proportion (%)
Contact	offering image	58	71.60	55	67.90
	demanding image	23	28.40	26	32.10
Distance	medium shot	25	50.00	35	44.30
	close-up shot	37	45.68	28	35.44
	long-shot	15	18.52	16	20.25
Perspective	horizontal angle	69	84.15	64	81.01
	vertical angle	13	15.85	15	18.99
Modality	high value modality	26	33.77	31	38.27
	medium value modality	31	40.26	30	37.04
	low value modality	20	25.97	20	24.69

As can be seen from the Table 5.2, in terms of contact, the proportion of offering image in China Daily and BBC News is higher than that of demanding image. This is because the purpose

of disaster news is to convey information rather than arouse sympathy to a certain extent. The proportion of offering image in China Daily is 3.7% higher than that in BBC News. Demanding images focus more on showing the interaction between viewer and the characters in the picture, attracting readers to participate in the visual interaction with the characters. The proportion of demanding image in BBC News is 3.7% higher than that in China Daily.

In terms of distance, most news reports use close-up and medium shooting techniques. The proportion of medium shoot in China Daily (50.00%) is higher than that in BBC News (44.30%). The proportion of close-up shoot in China Daily (45.68) is 10.24% higher than that in BBC News (35.44%). The proportion of long-shoot in both Chinese and British news is relatively low, but its proportion in BBC News (20.25%) is higher than that in China Daily (18.52%). Regarding perspective, in China Daily, the proportion of horizontal angle is 68.30% higher than that of vertical angle. In BBC News, the proportion of horizontal angle is 62.02% higher than that of vertical angle. The closer the shooting distance, the more vivid the central character and his facial expression will be, which is helpful in expressing the emotions of the person concerned; the farther the shooting distance, the smaller and more scattered the elements will be, which can easily make the reader lose the key points. The horizontal perspective is conducive to restoring the true position of the elements in the space and retaining their layers and three-dimensional shapes. However, vertical angle usually causes the original objects to overlap or be blocked, resulting in their deformation or loss of three-dimensional effect.

Regarding modality, the proportion of high-value modality in China Daily (33.77%) is lower than that in BBC News (38.27%). The proportion of medium-value modality in China Daily is higher than that in BBC News. The proportion of low-value modality in China Daily and BBC News is basically the same. The following photos of rescue operations during floods and tropical storms are selected and interpreted by the researcher in terms of the information and emotions conveyed by visual contact and shooting angles, etc.

5.2.1 Contact and distance

Regarding contact, if there is no eye contact between the characters in the image or between the viewer and the character, it is called “offering”, while if there is eye contact between the characters in the image or between the character and the viewer, it is called “demanding” (Kress & Van Leeuwen, 1996: 119). The distance of the shot affects the size of the elements in the picture in front of the viewer. The choice of distance can suggest different relationships between the participants represented and the viewers (ibid: 124). The medium shot uses a larger area in the picture to express the part of the subject, mainly showing the outline and posture of the character above the knee (Liu, 2007). Close-up shooting refers to presenting the more important part of the

subject than the medium shot, that is, the part above the person's chest (Liu, 2007). The long shot is the shot with the longest distance and the largest spatial range, and it can accommodate as much content as possible in the frame (Luo, 2011). The close shot represents the intimate relationship, the medium shot represents the social relationship, and the long shot represents the impersonal relationship (Kress & Van Leeuwen 1996:148). The Chinese and English news use shooting distances alternately, showing both the intimate relationship between the rescuers and the victims and the objective attitude of the observers of the event.

Figures 5.7 and 5.8 are demanding images. In Figure 5.7, the person in white clothes has eye contact with the rescued mother and daughter. When the viewer looks at this picture, he has eye contact with the rescuer in orange clothes behind the woman. Combining the mother's expression and the posture of holding the child tightly with both hands, viewers can sense her fear and worry. Both eye contacts can be considered as vectors, so this picture belongs to the demanding image.

Figure 5.8 shows a rescuer saving a child from a mudslide. The two people in the middle of the picture are looking at the child with very caring eyes, conveying their surprise. The child's eyes are looking at his family, and he seems to be about to cry, which deeply touches the viewer's sympathy. When we look at the picture, we interact with the smiling rescuers and feel their joy at this scene. The eyes of the others around all converge on the two central figures in the middle, conveying their concern and joy. When the viewer looks directly at the picture, he will feel sympathetic to people's misfortune and is delighted by their rescue. There are no vectors formed by gazes, tools, body movements, and arrows in Figure 5.9. It only presents the damaged house after the storm, so it is an offering image. This type of picture shows the detached attitude of the image user and objectively explains the situation, which does not leave a strong psychological impression in the viewer's mind.

Figures 5.7 and 5.8 can be regarded as close-up shots, making readers feel that they are closely related to the events in the picture. They basically present the part above the waist of the characters, indicating a relatively close psychological distance. Figure 5.9 is a long shot, which is often used to present the overall consequences of the disaster. The entire broken roof and the surrounding environment and even multiple people at the scene are included in the picture, and the characters are almost too small to be seen clearly. Long shot means that the viewer is independent of the picture or is psychologically far away from the events in the picture. This shooting distance usually includes more objects in the area, but it also has the disadvantages of scattered elements and the subject is not prominent.



Figure 5.7 Rescuers transfer flood-trapped people in Zhuozhou, Baoding city, North China's Hebei province.



Figure 5.8 Rescuers evacuate a baby after a mudslide hit Pingwu county, Sichuan province, on Tuesday.



Figure 5.9 Rescue operations continue in Karatsu.

5.2.2 Perspective and modality

Perspective is the angle of framing or shooting, which potentially causes the viewer's gaze to establish a positional relationship with the objects in the picture and affects the viewer's perception. The high angle forces the audience to look down at the participants, which may imply a sense of superiority over the things in the picture, while the low angle forces the audience to look up at them, which implies the audience's inferior position (Royce, 2007). However, the horizontal angle implies an equal relationship between the viewer and the characters in the picture (Kress & Van Leeuwen, 1996). Perspective reflects the intention of the photographer or image creator to guide the viewer to understand the image. For example, by showing the front, side or a specific angle of an object, it will leave different impressions on the viewer, such as positive, negative, ambiguous, or critical. Modality is a judgment of the authenticity of an image, and as details, sharpness, color, etc. reduce, the sense of modality will also decrease (ibid: 159).



Figure 5.10 President Chakwera visited some victims.



Figure 5.11 4 killed, 23 missing in Southwest China flash flood, mudslide



Figure 5.12 A close view at the Vistula River as water levels are at record low in Warsaw, Poland.

Figure 5.10 can be regarded as vertical perspective, because the line of sight of the official handing out the things and the woman receiving the things in the picture extend along the up and

down direction. The picture suggests that the official handing out the gift is looking down at the woman receiving it, which is not conducive to showing equal interpersonal relationships. The reader can realize that the man standing is in a superior position and is not approachable enough, while the woman in a lower position shows a respectful attitude and the two have an alienated status relationship. It makes the legs and head of the characters occupy a large space in the picture, so it is a medium shot. It is suggested that news adopt more horizontal perspectives when presenting character relationships to reduce unnecessary interpretations or misunderstandings for readers.

Figure 5.11 belongs to the horizontal perspective. On the one hand, the volunteers in the tent and the disaster victims are sitting and making eye contact. The picture shows their upper body from the waist up, which is the range of heights that are basically the same. On the other hand, from the positional relationship between the photographer and the people in the image, the image is shot from the horizontal perspective. Combining these two factors, we can determine that the image belongs to a horizontal perspective image. In terms of modality, Figure 5.11 belongs to a high value modality image. The image is composed of more than five colors, which forms a very obvious color difference. This helps the viewer quickly lock the information carried by the larger and more eye-catching area. Then, the image depicts the arms of the people, the light of the light bulb, and the two words “disaster relief” and “Sichuan” very realistically and clearly. The shadows of the people produced by the lighting are very realistic and three-dimensional, and the environmental clues represented by the tent are also very clear. Therefore, it belongs to a high-value modality image, which restores the efforts of the Chinese government departments to resettle the people in time during the disaster.

The perspective of Figure 5.12 is not clear because there are few reference objects in the picture. It is difficult for the researcher to judge the elements in the picture and their positional relationship with the photographer or viewer. It is a low value modality picture because there are only two colors in the picture. The boundary between the khaki cracked land and the light gray background in the upper part of the picture is also ambiguous. The picture lacks environmental clues such as grass or houses, that is, the details are very sparse and blurred. Moreover, there is no obvious light or shadow in the picture, which makes people feel depressed and uneasy. It highlights the drought caused by high temperature, which has caused people's production and life to fall into difficulties.

5.3 Compositional Meaning of Images

Kress & Van Leeuwen (1996:43) propose that any semiotic pattern must have the capacity to form a text, that is, a complex of signs that are internally related to each other and externally

consistent with the context. The compositional meaning of an image corresponds to the textual function of language, which refers to the layout of various elements in an image. Table 5.3 shows the proportion of each element in compositional meaning. Information in the center of the picture is usually more important. The position of the image is usually determined by the level of the subject in the picture, such as foreground and background. When an image is placed in the foreground or occupies a larger area of the picture than the reference object, it can be judged as the information that the news wants to highlight. The researcher labels the main image elements in the picture based on their position, relative size, color prominence, and the lines used to distinguish areas, thereby deriving the frequency and proportion shown in the table below.

Table 5.3. Frequency and proportion of elements in compositional meaning of images

Compositional Meaning		China Daily		BBC News	
Elements	Categories	Frequency (times)	Proportion (%)	Frequency (times)	Proportion (%)
Information Value	center and margin	43	52.44	38	48.10
	left and right	22	26.83	25	31.65
	top and bottom	17	20.73	16	20.25
Saliency	position	36	40.45	39	46.99
	color saturation	28	31.46	30	36.14
	relative size	25	28.09	14	16.87
Framing	dividing line	34	16.59	13	7.34

As can be seen from the Table 5.3, in China Daily and BBC News, information value and color saliency are the most important in highlighting the compositional meaning of images. In terms of information value, the center and margin layout have the highest proportion, which is 52.44% and 48.10% respectively. The proportion of left-right layout pictures in China Daily is 4.82% lower than that in BBC News. Top and bottom layout images account for the lowest proportion, 20.73% and 20.25% in China Daily and BBC News respectively. In terms of saliency factors, the position factor accounts for 40.45% and 46.99% in China Daily and BBC News respectively, making the greatest contribution to saliency. The color saturation in China Daily (31.46%) is lower than that in BBC News (36.14%). The framing is shown by dividing lines, and its proportion in China Daily is much higher than that in BBC News, 16.59% and 7.34% respectively.

5.3.1 Information value and saliency

The color, shape, combination, size and other designs of symbols have different effects on the meaning of the image. The compositional meaning refers to the layout of various elements in the picture, which has an important impact on the importance and attractiveness of information. The left side of the image is assumed to be “known” information, and the right side is regarded as

“new” information, as not known to the addressee and not shared by the social group (Kress, 2009:92). People are used to browsing images from left to right, so that the brain processes the information on the left first, and then deepens the perception through the content on the right. The contrast between dark and light colors can create a strong visual impact. The photographer cleverly uses techniques to arrange elements at different levels and the size of objects to influence the importance of information.



Figure 5.13 A fire crew member fights part of the Hillville bushfire south of Taree in New South Wales, Australia, on Tuesday.



Figure 5.14 Members of Chongqing Blue Sky Rescue Team carry out the search and rescue mission in quake-hit Malatya.



Figure 5.15 Firefighters were helped by wetter weather towards the end of the week.

In terms of information value, Figure 5.13 and Figure 5.15 belong to left-right layout. In Figure 5.13, the firefighter is located in the lower left corner of the entire picture in which he uses water cannons to put out fires as an active participant. The flames occupy two-thirds of the entire picture from the center to the right. In Figure 5.15, the burned building occupies two-thirds of the picture on the left, and the firefighter stands in the right third of the picture. He does not have any firefighting tools in his hands and his feet are together. His limbs are in a state of stillness rather than movement. Combined with the text below, we can know that the humid weather has weakened the fire. Figure 5.13 is more informative than Figure 5.15 because Figure 5.13 suggests that the firefighter is trying to put out the raging fire with a useful tool, while the firefighter in Figure 5.15 is more like a silent bystander. Figure 5.14 belongs to the center-margin layout. Four rescuers are located in the center of the picture and a jumble of collapsed houses serves as the background. This image highlights the efforts of the China Blue Sky Rescue Team in providing aid overseas.

In terms of salience, Figure 5.13 shows the brave image of firefighters through relative size. The firefighters here occupy a very small part of the lower right corner of the picture, while the entire rising flames and the water sprayed by the fire-fighting equipment occupy a larger area. This contrast highlights the horror of natural disasters and the insignificance of humans in the face of

natural disasters. In the footage, the firefighter is not afraid of the raging fire, and he operates fire-fighting equipment calmly and bravely, warning people to pay attention to forest fire prevention.

Figure 5.14 highlights the most important information of the picture through position and color saturation. The three members of the Blue Sky Rescue Team are placed in the foreground of the picture, occupying a larger proportion of the center of the picture. Moreover, the camera focuses on them looking carefully at the equipment in their hands, allowing readers to associate them with their detection and rescue operations. The importance of the background, a jumble of collapsed houses, is greatly diminished. When looking directly at this image, the viewer first focuses on the information in the center, which is the rescue team. It can be inferred that it downplays the collapsed buildings after the earthquake and emphasizes the efforts of the Chinese rescue team. The color saturation in it is very obvious, such as grey collapsed buildings, rescue workers in blue, white snow. The bright yellow letters “BSR” on the rescuers’ backs are the abbreviation of “Blue Sky Rescue”. The contrast between warm and cold colors makes the team name more eye-catching. Images with such high color saturation are seen as more credible.

From the perspective of position, Figure 5.15 highlights the dramatic impact of the fire through the background and foreground. When the audience looks directly at the picture, they tend to first focus on the larger background, which is composed of gloomy weather, huge forests and burned buildings. Secondly, a single firefighter with no facial details is placed in the right foreground, with his back to the viewer and lacking eye contact with them. The whole picture has a blurred contrast between light and dark and simple environment details, which can be judged as a low-value image. This implies a more negative and sad tone. Frankly speaking, this gloomy atmosphere is not conducive to conveying more positive emotions in the context of the disaster, but will make readers feel depressed and frustrated.

5.3.2 Framing

Some news illustrations have elements similar to lines that divide them into areas with different characteristics. The dividing line can be horizontal, vertical, explicit, implicit, curved, straight, etc. Moreover, it can be a line or an object, such as a door frame and a fence. Although the role and status of the frame are not very prominent compared to other elements, it can make the contrast of information in each area of the picture more obvious. As far as the framing is concerned, there is a white dividing line between the upper and lower parts in Figure 5.16. It divides the entire incident into two frames, the tidy shops before the disaster and the mess after the disaster. The upper left corner of the area above the dividing line shows Jan 2024, which is the normal local situation before the mudslide occurs. The upper left corner of the lower half of the picture is marked with 12 May 2024, when the disaster has already occurred and the original area

has become a mess. The dividing line highlights the situation before and after the disaster, providing the audience with a very intuitive visual comparison.

Figure 5.17 shows that the epicenter of the earthquake is located about 68 kilometers away from the coastal town of Ferndale. The horizontal latitude lines and vertical longitude lines in the image intersect with each other, which allows the viewer to quickly determine the latitude and longitude of the area affected by the earthquake, and thus determine its location. On the right side of the image, the coastline near the center of the earthquake, Ferndale, represented by the five-pointed star, divides the land and the sea. According to the annotations below the image, the dark yellow and light yellow areas on the east side of the coastline form an implicit dividing line, representing earthquakes of magnitude six and seven respectively. Here, the dividing line plays a role in clarifying the scope and intensity of the earthquake. It can be seen that Fortuna and Eureka are the areas with the highest earthquake intensity, reaching more than six.



Figure 5.16 Destroyed houses in pandai sikek, Tanah Datar Regency, West Sumatera

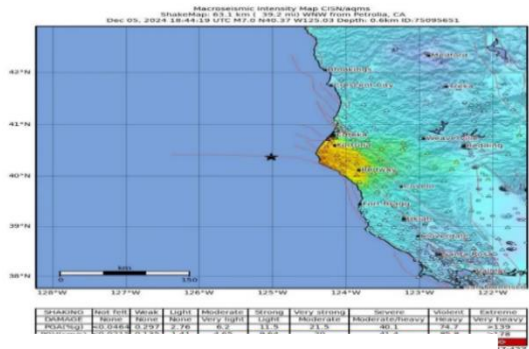


Figure 5.17 Macroseismic Intensity Map

In summary, this section introduces the concepts of representational meaning, interpersonal meaning, and compositional meaning, as well as their symbolic representation forms, obtaining the frequency and proportion of symbolic elements through annotation. In particular, the case analysis interprets the influence of the combined components, arrangement order, shooting angle, etc. of images in disaster news on their meaning from macro to micro. Strategies such as close-up shooting, eye-level perspective, and visual interaction can trigger the psychological response of viewers to perceive images. The vectors in the picture, as an important medium to attract the audience’s attention, usually guide the viewer’s eyesight to the rescue measures and the damage at the accident site.

6. Conclusion

This chapter discusses the findings of the research firstly, involving the lexical and grammatical characteristics of natural disaster news reflected in data and text analysis and its significance in the context of disaster. Secondly, it summarizes the implications brought by the study from multiple perspectives, especially focusing on the improvement of the professional quality of journalists and the framework of news reports. Finally, from the perspective of corpus collection and analysis framework, it summarizes the shortcomings of existing research and future improvement strategies.

6.1 Findings and Implications

This study applied multimodal discourse analysis framework to examine the proportion of language resources and image resources in natural disaster news of China Daily and BBC News and their emergency management functions. It focuses on exploring how media workers use language and images to convey their views and attitudes on disasters. Then, by showing the collocation of verbs and objects, and the distribution of theme-rheme, etc, it reveals how news changes the information structure, thereby highlighting key information. Moreover, by examining the role of image symbols in reproducing the disaster scene and triggering a sense of care, it provides suggestions for improving the attractiveness and comprehensibility of news.

Research shows that there is a certain interactive mechanism between language and image. Language is dominant, while images can highlight, supplement or confirm the content described by language, and also play an important role in attracting viewers' attention. According to systemic functional grammar and visual grammar, the various elements of language resources and symbolic resources play different degrees of roles in describing facts, emotional connection and structural layout.

There are certain differences in the proportion of various symbolic elements in China Daily and BBC News, but they are generally similar. Taking transitivity as an example, material process accounts for the highest proportion in the two corpora, reaching 71.77% and 65.02% respectively. Existential process is usually related to static and stable events within a certain time and space, while disaster news is mainly about sudden and changing situations. Behavioral process is usually about personal physiological activities and is not closely related to the information needs of the majority of news readers. Therefore, in the two corpora, existential process and behavioral process account for the lowest proportion.

The differences in news strategy and content between China and Britain and their causes are briefly summarized here. China Daily tends to use declarative sentences as news headlines, which

can summarize the general situation or main information of natural disasters. However, in BBC News, many bold or italicized news headlines are interrogative and exclamatory sentences, which are more likely to trigger readers' thinking and psychological reactions. Secondly, China Daily emphasizes the rescue forces and post-disaster recovery of various social departments in disaster reports, while BBC News prefers to highlight the severity of the disaster and the adverse effects it brings. In addition, Chinese media pay more attention to post-disaster reconstruction and people's safety, while British media focus more on the disaster itself or the dedication of an individual. The differences in reporting tendencies are influenced by the cultural differences between China and the West. China Daily usually follows the pace of the Chinese government closely. The reported content is in line with the government's call and does its best to avoid causing panic or public opinion outbursts in the context of disasters. The reports of BBC News have the color of Western liberal thought and individualism, paying more attention to the responsibility of supervision. Therefore, sometimes they condemn the inefficiency or inaction of the official rescue.

Natural disaster news plays emergency management functions such as information transmission, emotional comfort, confidence motivation, and public opinion guidance. Moreover, in response to the problems emerging in natural disaster news, the writer also provides corresponding suggestions. First, in terms of language resources, a small part of natural disaster news provides popular science content mainly on the conditions of disaster process and the causes of casualties, but provides few escape and rescue measures for people. Secondly, the frequency of high-value modal words in disaster accident reports is low, although news reports are objective and rigorous. However, the appropriate use of high-value modal words can show the official agencies' firm determination and measures for disaster relief, so that the affected people have more confidence and hope to overcome difficulties. In terms of image modality, many media reports contain images with complex elements and blurry quality, which cannot bring a good impression to readers. Moreover, in many news pictures, the rescuers are facing away from the readers and the tones of image are gray, which easily creates a sad and frustrating atmosphere. This is not conducive to rallying people's hearts and stabilizing society under the background of disasters.

Therefore, it is recommended that news reports should popularize emergency knowledge and provide escape and rescue strategies in the form of short videos and comics based on the type of disaster. Firstly, media organizations can establish a team specifically responsible for disaster reporting to learn safety knowledge and produce concise and vivid emergency rescue publicity works. These promotional works with interesting forms are conducive to improving people's safety knowledge and rescue capabilities and reflect the social service value of news. Secondly, news reports can use more transitional markers and frame markers to enhance the coherence and logic of the text, highlighting the key points. These indicative words help readers establish a

framework for understanding, thereby effectively guiding them to quickly obtain clear and accurate information they need. Moreover, they can appropriately increase the number of pictures of rescuers facing the readers at work, and improve the brightness and color contrast of the pictures. Finally, media organizations can set up interactive columns or hotlines for readers in disaster-stricken areas, listen to their voices and offer encouragement and assistance full of humanistic care. This will help to clearly display things, encourage the affected people, and establish the image of responsible national workers.

6.2 Limitations and Suggestions

In the above content, we have sorted out the language differences and similarities between Chinese and English news and the problems in their reporting, and proposed solutions. However, there are still some deficiencies in corpus collection and text analysis.

The corpus collected in this study is limited to news released by representative media in China and the United Kingdom, so the diversity of the corpus is insufficient. This leads to the conclusions drawn in this study being weak in persuasiveness and practical application value. Videos record disaster events more realistically and comprehensively, and have a more direct propaganda and warning effect. In this fast-paced era, videos are becoming more and more popular among the public. Different melodies in videos have different psychological implications for the audience, such as inspiring, depressing, and regretful, which will affect the audience's emotions and are therefore worth exploring. However, this study lacks research on the video and music modes embedded in news.

In the text analysis stage, the theories used are systemic functional grammar and visual grammar. The writer focuses on the emotions and attitudes of the victims and news readers, but ignores the cognitive and interactive factors of the two subjects. Moreover, the images and texts of news have relationships of explanation, proof, supplementation, and multiple spatial layouts, and their meaning expression effects should be studied.

In terms of solutions, the researcher will collect disaster reports from media in different countries and disaster reports released by official media and local media to ensure the diversity of the corpus in the future. In addition, with the help of the video annotation software Elan, the relationship between the video content and the pitch, rhythm and melody of the accompanying music will be analyzed to deepen the understanding of multimodal synergy.

The news creator's thinking and social and cultural factors will affect the discourse construction of disaster news, which implies that certain values will invisibly affect the readers' psychology. Metaphor, as a reflection of thinking mode, is an effective entry point for studying the implicit meaning of news. Therefore, future research will draw on theories such as critical

metaphor analysis to explore factors such as cognition and interaction reflected in disaster news. Questionnaire survey will be added to understand the perception of disaster metaphors by audiences in different cultural background, so as to obtain more practical inspiration. The writer will also study images and text as a whole to explore the impact of the relationship between images and text on the expression effect.

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